

Of Fathers and Farms

How Intergenerational Persistence Shapes Structural Transformation

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Abstract. How much faster could the United States have industrialized if new workers were not more likely to work in their parents' sectors? To study this question, I develop a two-sector general equilibrium Roy model where workers are either more productive or otherwise more attached to their parents' sectors. I show that both mechanisms meaningfully affect the static and dynamic sectoral reallocation of workers in response to labor demand shocks. Taking the model to linked father-son records in census microdata, I find that both mechanisms are quantitatively important sources of intergenerational persistence in sector choice. Counterfactual simulations reveal that eliminating intergenerational persistence altogether increases U.S. industrialization between 1880 and 1940 by 13.3 percent, but this is almost entirely explained by the inherited productivity mechanism.

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Introduction

Intergenerational mobility is a major topic concerning both economists and policymakers, largely due to its implications for both inequality and individual opportunity. However, when individual workers are systematically drawn toward their parents' specific occupation or sector, this persistence also determines the aggregate supply of labor. In this paper, I ask how intergenerational persistence in sector choice influences an economy's structural transformation.

I build on a growing literature that emphasizes the importance of new generations of workers for structural transformation ([Hobijn, Schoellman and Vindas, 2018](#); [Porzio, Rossi and Santangelo, 2022](#)). Without an employment history of their own anchoring them to a particular sector, new workers are commonly viewed as much more responsive to changing economic conditions than incumbent workers. Yet, with workers disproportionately drawn toward their parents' occupations, new workers may be less responsive than conventionally assumed. In this paper, I argue that this is one way in which intergenerational persistence reshapes an economy's structural transformation. In addition, I show that intergenerational persistence also transmits worker reallocation forward to future generations.

The historical episode of U.S. structural transformation in the first half of the 20th century provides an ideal setting for this analysis. From 1880 to 1940, the share of U.S. workers in agriculture fell from 61 percent to 23 percent with new workers accounting for the vast majority of this decline.¹ Using links across and within the full-count decennial censuses available from IPUMS ([Ruggles et al., 2024](#); [Helgertz et al., 2024](#)), I observe new workers' sector choices together with the sector in which their fathers worked. In 1880, more than four-in-five workers whose father worked in agriculture (i.e., "ag-born") worked in agriculture themselves, as did fewer than one-in-five workers whose father worked in nonagriculture (i.e., "nonag-born"). This persistence cannot be trivially explained by basic differences in childhood school attendance or local geography. Because historical microdata cannot fully unpack the deeper unobservable forces driving this association—such as unmeasured human capital, specific skills, or wealth transfers—I develop a structural model to map these micro-forces into macroeconomic wedges.

I build on a two-sector partial equilibrium [Roy \(1951\)](#)-type model in which I model two sources of intergenerational persistence as wedges.² I consider two

¹ According to [Hobijn, Schoellman and Vindas \(2018\)](#), new workers accounted for roughly 70 percent of that reallocation.

² [Hsieh et al. \(2019\)](#) use a similar approach to study the allocation of talent in U.S. labor market,

broad explanations for why the sons of U.S. farmers are disproportionately attached to agriculture compared to the sons of nonfarmers. On one hand, workers may inherit their parents' comparative advantage: ag-born workers are more skilled in agriculture and nonag-born workers are more skilled in nonagriculture. I call this *inherited productivity*. On the other hand, the sons of farmers may find it difficult or otherwise unappealing to pursue nonagricultural work due to cultural or institutional barriers, financial or information frictions, preferences, etc. I call this *inherited immobility*. Though broadly defined, I find that these two mechanisms have sharply different implications for structural transformation, especially due to their different effects on the dynamic transmission of past worker reallocation to future generations.

To identify these wedges, I leverage equilibrium conditions of the model to map observed sector choices and earnings into the underlying productivity and immobility wedges. With inherited productivity, the model predicts that ag-born workers are more skilled on average than nonag-born workers in agriculture, and vice versa. This productivity difference is also reflected in the average earnings difference between ag-born and nonag-born workers in each sector. As documented by [Lagakos and Waugh \(2013\)](#), the average productivity and hence the average earnings of ag-born versus nonag-born workers in a given sector will also reflect gains from self-selection. As such, I can only identify the inherited productivity wedge from earnings after accounting for selection. Finally, the inherited immobility wedge is identified by the residual intergenerational persistence in sector choice not explained by inherited productivity.

I estimate the wedges by taking the model's identifying equilibrium conditions to historical U.S. decennial census microdata ([Ruggles et al., 2024](#)). I focus on 20–29 year-old U.S.-born men in 1940 who I can link to their childhood selves in 1930 using the IPUMS Multigenerational Longitudinal Panel ([Helgertz et al., 2024](#)). I use this linked data to observe the sector in which each worker's father worked in 1930, as well as the sector in which the worker himself works in 1940. Together, these linked father–son records allow me to measure the intergenerational persistence in sector choice and the average earnings of ag-born and nonag-born workers in each sector. After accounting for the gains from selection, I estimate that ag-born workers in agriculture are paid 42 percent more than nonag-born workers, while nonag-born workers in nonagriculture are paid 81 percent more than ag-born workers. Through the lens of the model, this implies that workers are 42 and

especially the role of racial or gender discrimination.

81 percent more productive in their parents' sector, respectively. Hence, inherited productivity accounts for about two-thirds of U.S. intergenerational persistence in 1940. I attribute the remaining one-third of intergenerational persistence to inherited immobility. Viewed through the lens of the model, this wedge suggests that the sons of farmers sort as if they earn 40 percent less in nonagriculture.

In order to shed light on intergenerational persistence's long-run macroeconomic consequences, I embed the partial equilibrium framework into a general equilibrium environment with standard drivers of structural transformation. Technological progress generates shifts in relative labor demand through a combination of relative price and income effects, which incentivizes new workers to allocate differently than their parents' generation. I show that new workers' sector choices are less elastic to these labor demand shocks driving structural transformation when intergenerational persistence is present. In particular, the intergenerational persistence observed in the 1880 U.S. economy implies that new workers are roughly 45 percent less responsive to contemporaneous changes in relative wages than they would be in the benchmark economy. In the general equilibrium setting, worker reallocation induced by a labor demand shock generates additional wage responses that slightly offset the initial shock. I show that inherited immobility amplifies this feedback more than inherited productivity; hence, the contemporaneous reallocation of workers is smallest when intergenerational persistence is driven by inherited immobility.

In the model, workers solve a static problem and live for a single period; nevertheless, there is a dynamic transmission of past worker reallocation to future generations due to intergenerational persistence. This unfolds in two parts. If workers leave agriculture in a given generation, the next generation contains fewer ag-born workers. Since ag-born workers are more likely to work in agriculture than nonag-born workers, past reallocation makes the next generation of workers immediately less likely to work in agriculture as well. The second part of the dynamic transmission is more subtle and depends on the underlying source of intergenerational persistence. With fewer ag-born workers in the next generation, the total skill supplied to each sector is tilted away from agriculture and toward nonagriculture. That shift induces a change in relative wages that raises the incentive for workers to pursue agricultural work. With inherited immobility alone, the reallocation out of agriculture that the economy missed out on in the current generation is partially recovered through additional departure out of agriculture in the next generation. With inherited productivity, the shift in total skill supply is much

more pronounced and can generate a wage response large enough to not only offset the first channel but to actually reverse it. Hence, with inherited productivity the fewer workers leaving agriculture in the current generation is reinforced by workers returning to agriculture in the next generation.

I calibrate the model to match the United States' growth and transformation between 1880 and 1940. Consistent with past work, I infer especially rapid technological progress in the nonagricultural sector, which pulls workers out of agriculture primarily through income effects ([Dennis and İřcan, 2009](#); [Alvarez-Cuadrado and Poschke, 2011](#)). I simulate counterfactuals in which the U.S. economy starts from its observed 1880 allocation of workers and experiences the same technological growth and other historical forces through 1940, but under different regimes of intergenerational persistence. I find that removing intergenerational persistence altogether leads to 5.1 percentage points (pp.) more reallocation of workers out of agriculture by 1940, equal to 13.3 percent of the observed reallocation. While I find that intergenerational persistence as a whole was a substantial drag on U.S. industrialization, the two underlying mechanisms do not contribute equally to this. If I remove only inherited immobility, U.S. industrialization increases by only 1.1 pp. (i.e., 2.9 percent of the observed reallocation). However removing only inherited productivity increases U.S. industrialization by 5.5 pp. (i.e., 14.4 percent of the observed reallocation). That large magnitude difference reflects the different ways in which inherited productivity and inherited immobility shape the dynamic transmission of past worker reallocation to future generations.

Relevant Literature A large literature explains structural transformation through changes in sectoral productivity, relative prices, and the composition of demand (see [Herrendorf, Rogerson and Valentinyi, 2014](#), for a survey).³ My paper instead focuses on how labor supply adjusts across sectors, building on research that incorporates labor-supply frictions, worker heterogeneity, and cohort replacement into structural transformation ([Kim and Topel, 1995](#); [Lee and Wolpin, 2006](#); [Lagakos and Waugh, 2013](#); [Herrendorf and Schoellman, 2018](#)). [Hobijn, Schoellman and Vindas \(2018\)](#) and [Porzio, Rossi and Santangelo \(2022\)](#) show that new workers accounted for the majority of the reallocation of workers out of agriculture in the U.S. and globally, respectively. My paper re-examines the role of new workers

³Canonical contributions include [Kongsamut, Rebelo and Xie \(2001\)](#); [Gollin, Parente and Rogerson \(2002\)](#); [Ngai and Pissarides \(2007\)](#); [Restuccia, Yang and Zhu \(2008\)](#); [Rogerson \(2008\)](#); [Duarte and Restuccia \(2010\)](#); [Buera and Kaboski \(2012\)](#).

in structural transformation when their sector choices are connected across generations to their parents'. The most closely related work is [Hayashi and Prescott \(2008\)](#), who consider institutional norms forcing farmers' sons to take over the family farm as an impediment to structural transformation in prewar Japan. My analysis also allows persistence to arise from either inherited productivity or inherited immobility, which I discipline empirically using linked father–son census records. Moreover, I show that these two paradigms of intergenerational persistence have sharply different implications for structural transformation.

Another large but distinct literature has studied intergenerational persistence in occupational choice ([Rogoff, 1950](#); [Blau and Duncan, 1967](#); [Long and Ferrie, 2013](#); [Haack and Laliberté, 2025](#)).⁴ Historical work using linked censuses traces this persistence across periods and generations ([Olivetti and Paserman, 2015](#); [Feigenbaum, 2018](#); [Long and Ferrie, 2018](#); [Ferrie, Massey and Rothbaum, 2021](#); [Song et al., 2020](#); [Braun and Stuhler, 2018](#)). Using full-count U.S. censuses, [Liu \(2026\)](#) provides related empirical evidence that the Dust Bowl shifted the occupational choices of children from farming households, particularly those exposed at younger ages. Related evidence shows that families transmit occupation-specific human capital, access to employers and professions, information, and exposure ([Laband and Lentz, 1983](#); [Corak and Piraino, 2011](#); [Aina and Nicoletti, 2018](#); [Bell et al., 2019](#); [Greenberg et al., 2024](#)). In agriculture specifically, [Haws, Just and Price \(2025\)](#) use linked censuses to document population-level patterns of family-farm succession, highlighting the direct intergenerational transmission of farms as a distinctive source of occupational persistence. Most papers have focused on measuring intergenerational persistence, but a nascent strand examines its macroeconomic consequences. For example, [Lo Bello and Morchio \(2022\)](#) use a search framework to decompose intergenerational persistence in the contemporary U.K. and study its implications for the static allocation of talent. Likewise, [Almgren, Kramer and Sigurdsson \(2025\)](#) use a Roy-type framework to analyze whether intergenerational occupation persistence in contemporary Sweden reflects a misallocation of talent. In contrast, I study how intergenerational persistence shapes the dynamic evolution of worker allocations in response to changing economic fundamentals.

⁴The broader economic literature typically models persistence through parental investments and the transmission of human capital; canonical contributions include [Becker and Tomes \(1979, 1986\)](#); [Solon \(1999\)](#), while [Black and Devereux \(2011\)](#) provide a survey. [Abramitzky, Boustan and Matiasvili \(2025\)](#) provide a recent survey focused on intergenerational occupational mobility.

Roadmap Section 1 presents the motivating evidence on intergenerational persistence in the U.S. between 1880 and 1940. Section 2 develops a partial equilibrium Roy-type model of sector choice with intergenerational persistence and uses it to measure inherited productivity and immobility wedges. Section 3 embeds the partial equilibrium model into a general equilibrium environment and studies how intergenerational persistence shapes the dynamic evolution of worker allocations in response to changing economic fundamentals. Section 4 calibrates the model to the U.S. economy and quantifies the macroeconomic consequences of intergenerational persistence for its structural transformation in the first half of the 20th century. Section 5 concludes.

1. Measuring Historical U.S. Intergenerational Persistence

As mentioned above, [Hobijn, Schoellman and Vindas \(2018\)](#) and [Porzio, Rossi and Santangelo \(2022\)](#) document that new workers are essential for labor reallocation out of agriculture. I build on this insight by examining how the sectoral choice of new workers is influenced by their parents' sector. I document that the early-20th-century U.S. economy is characterized by very high intergenerational persistence in sectoral employment. Specifically, the sons of farmers (i.e., "ag-born workers") are substantially more likely to work in agriculture than the sons of nonfarmers (i.e., "nonag-born workers"). I show that this persistence is robustly measured within local labor markets and after conditioning on schooling. Moreover, it is much higher in agriculture than in other sectors.

To document these facts, I use full-count U.S. decennial census microdata for 1860–1940 available from [Ruggles et al. \(2024\)](#). I restrict the data to U.S.-born men aged 20–29 who are working and whose industry is observed in the census. Specifically, I classify workers as agricultural if their industry is classified by IPUMS as agriculture⁵ and nonagricultural otherwise. In order to measure intergenerational persistence, I must observe each worker's parental sector. To this end, I use the IPUMS Multigenerational Longitudinal Panel ([Helgertz et al., 2024](#)) to link new entrants to their fathers in earlier censuses. This approach builds on a growing economic-history literature that uses linked historical records to measure intergenerational mobility in the United States ([Long and Ferrie, 2013](#); [Abramitzky, Boustan and Eriksson, 2014](#); [Olivetti and Paserman, 2015](#); [Feigenbaum, 2018](#); [Song et al.,](#)

⁵IND1950 codes 100–199.

2020).⁶

In Figure 1, I plot the agricultural employment share of U.S.-born men aged 20–29 in each census year. This allocation of new workers across sectors exhibits a significant downward trend, falling by roughly 38 percentage points over the period.

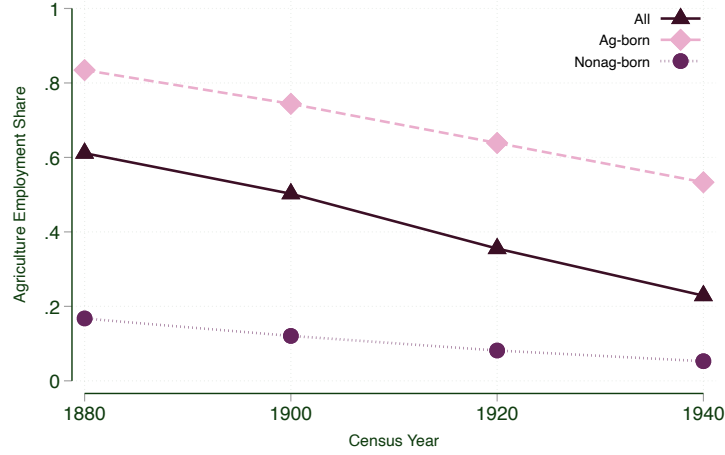


Figure 1: Industrialization of New U.S. Workers, 1880–1940

Notes: The figure plots agricultural employment shares among linked U.S.-born employed men ages 20–29 with observed own and paternal industry. Ag-born workers have fathers employed in agriculture in the earlier childhood census, while nonag-born workers have fathers employed outside agriculture. Agriculture is defined by IND1950 codes 100–199, and the “All” series pools ag-born and nonag-born workers in the linked sample.

Critically, the sons of farmers and the sons of nonfarmers allocate themselves very differently across sectors. In 1880, over 80 percent of ag-born workers are employed in agriculture compared to less than 20 percent of nonag-born workers. Consider the following linear probability model estimated among individuals i in census year t :

$$\mathbb{1}\{\text{Sector} = \text{Agriculture}\}_{i,t} = \beta_0 + \beta_1 \cdot \mathbb{1}\{\text{Father's sector} = \text{Agriculture}\}_{i,t} + \varepsilon_{i,t}$$

Without any additional covariates, the estimated coefficient β_1 exactly equals the difference in agricultural employment rates between ag-born and nonag-born workers as depicted in Figure 1. That unconditional gap is 66.7 percentage points as noted in column 1 of Table 1. Next, I establish that the high degree of intergenera-

⁶Specifically, I use the MLP to link individuals to their own childhood records in earlier censuses, and then use the intra-census household links to identify their fathers and their fathers' industries.

tional sector persistence in the U.S. is not trivially explained by access to schooling or geography.

Table 1: Likelihood of Agricultural Employment for New Workers, 1880

	(1)	(2)	(3)
Father in agriculture	0.667*** (0.0008)	0.665*** (0.0009)	0.548*** (0.0011)
Schooling fixed effects	No	Yes	Yes
Schooling $\times$ CZ fixed effects	No	No	Yes
Observations	855,723	855,723	855,723
R^2	0.417	0.418	0.470

Notes: The dependent variable is an indicator for whether a son works in agriculture, and the reported coefficient corresponds to an indicator for whether his father works in agriculture. Column 1 includes no additional controls, column 2 includes childhood schooling fixed effects, and column 3 includes childhood-schooling-by-commuting-zone fixed effects. The sample consists of linked U.S.-born employed men ages 20–29 in 1880 with observed own and paternal industry, and agriculture is defined by IND1950 codes 100–199. Robust standard errors are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

A large literature emphasizes the role of schooling for structural transformation (e.g., [Caselli and Coleman II \(2001\)](#); [Porzio, Rossi and Santangelo \(2022\)](#)). Thus, an immediate concern is whether ag-born workers may be more likely to pursue agricultural work only because they have systematically different access to schooling than nonag-born workers. I test for this in column 2 of Table 1 by including an indicator for whether the worker was attending school when observed during their childhood. The schooling-conditional gap in agricultural employment between ag-born and nonag-born workers is 65.9 percentage points, effectively unchanged from the unconditional gap.

Another concern is that the aggregate U.S. economy may comprise two dramatically different types of labor markets: agrarian hinterlands and industrialized cities. To the extent that ag-born workers are born in the hinterland while nonag-born workers are born in the city, the observed intergenerational persistence may be explained by workers’ inability or unwillingness to move. Recent work by [Eckert, Juneau and Peters \(2023\)](#) shows that U.S. industrialization from 1880 to 1940 was highly localized, with 60–90 percent of reallocation occurring within commuting zones. Likewise, I include commuting-zone⁷ fixed effects in column 3 of Table

⁷I define commuting zones using the crosswalks from [Eckert et al. \(2020\)](#).

1 and find that the allocation gap between ag-born and nonag-born workers measured within local labor markets is only about 18 percent smaller than the unconditional gap.

Table 2: Intergenerational Persistence Across Sectors for New Workers, 1880

	(1)	(2)	(3)
	Ag vs. manu.	Ag vs. serv.	Manu. vs. serv.
Father in agriculture	0.552*** (0.0016)	0.548*** (0.0012)	
Father in manufacturing			0.271*** (0.0019)
Outcome: son in agriculture	Yes	Yes	No
Outcome: son in manufacturing	No	No	Yes
Schooling $\times$ CZ fixed effects	Yes	Yes	Yes
Observations	655,141	765,212	291,093
R^2	0.348	0.424	0.137
Implied odds ratio	12.29	11.92	3.56

Notes: The dependent variable in columns 1 and 2 is an indicator for whether the son works in agriculture, while the dependent variable in column 3 is an indicator for whether the son works in manufacturing. Columns 1 and 2 report the coefficient on an indicator for whether the father works in agriculture and restrict the comparison group to fathers in manufacturing and services, respectively. Column 3 restricts the sample to sons of nonagricultural fathers and reports the coefficient on an indicator for whether the father works in manufacturing. Manufacturing is defined by IND1950 codes 300–499, while services include all other nonagricultural industries. All specifications include childhood-schooling-by-commuting-zone fixed effects, and the sample, odds-ratio construction, and inference conventions follow Table 1.

Finally, I gauge whether the unconditional gap is driven by ag-born workers' attachment to agriculture or instead reflects intergenerational persistence across sectors more generally. Columns 1 and 2 of Table 2 compare ag-born workers with workers whose fathers were employed in manufacturing and services, respectively. The estimated coefficients are effectively the same as in the full sample reported in column 3 of Table 1. Thus, the agricultural employment gap is not driven by the particular mix of parental industries represented among nonag-born workers.

Column 3 of Table 2 instead restricts the sample to workers with nonagricultural fathers and compares the manufacturing employment shares of workers with manufacturing fathers and those with service-sector fathers. Rather than directly

comparing the estimated coefficients, I convert the average counterfactual probabilities from each linear probability model into the corresponding odds ratio, reported in the final row of Table 2.⁸ After conditioning on schooling and location, the sons of farmers are roughly 12 times more likely to work in agriculture than the sons of nonfarmers, and this is approximately the same whether the comparison group is restricted to fathers in manufacturing or services. Meanwhile, the sons of manufacturers are only 3.5 times more likely to work in manufacturing than the sons of service workers. Thus, while there is intergenerational persistence across all sectors, it is dramatically higher in agriculture than in other sectors.

Discussion These findings establish intergenerational persistence in sectoral employment as a quantitatively important feature of the U.S. economy during this period of structural transformation. Yet the historical microdata alone provide limited leverage for identifying the forces underlying this persistence: schooling and local labor-market conditions account for little of the observed relationship between fathers' and sons' sectors. Broadly, the sons of farmers may remain in agriculture because they are more skilled in that sector or because they otherwise face lower incentives to pursue nonagricultural work. The next section formalizes these two paradigms as inherited productivity and immobility wedges. Combining the model's structure with data on workers' sectoral choices and earnings allows me to identify these wedges and, in subsequent sections, quantify how intergenerational persistence shaped the U.S. economy's transition out of agriculture.

2. Modeling Sector Choice with Intergenerational Persistence

I incorporate intergenerational persistence into a Roy-type model of comparative advantage and self-selection (Roy, 1951; Heckman and Sedlacek, 1985; Heckman and Honoré, 1990; Sattinger, 1993), adapted to the structural-transformation setting as in Lagakos and Waugh (2013) and to a wedges approach as in Hsieh et al. (2019). Workers have heterogeneous skills for two sectors of production, agriculture and nonagriculture. They choose their sector to maximize indirect util-

⁸The odds ratio measures the relative likelihood of a son working in sector j given that his father works in sector j versus given that his father works outside sector j . It is computed as:

$$\frac{\Pr(\text{Works in Sector } j \mid \text{Father works in Sector } j)}{\Pr(\text{Works in Sector } j \mid \text{Father works outside Sector } j)} \bigg/ \frac{\Pr(\text{Works outside Sector } j \mid \text{Father works in Sector } j)}{\Pr(\text{Works outside Sector } j \mid \text{Father works outside Sector } j)}$$

ity, which depends on their earnings and sector-specific utility shifters.⁹ I extend the standard framework by incorporating two sources of intergenerational persistence: inherited productivity and inherited immobility.¹⁰ I model inherited productivity as a wedge that rescales workers' skills in their parents' sectors, while inherited immobility is a wedge that reduces ag-born workers' indirect utility from working in nonagriculture.

By assuming that workers' skills are Fréchet distributed, I can derive closed-form expressions for the fractions of ag-born and nonag-born workers who choose each sector, as well as the average skill they supply to each sector. I leverage these expressions to identify the model's four labor supply parameters—the Fréchet shape parameter, the inherited productivity wedges, and the inherited immobility wedge—from four moments concerning workers' income and sectoral allocations. Taking the model to data, I use the linked father–son census records described previously to infer these parameters.

2.1. Environment

There is a unit mass of households indexed by i and two sectors of production indexed by $j \in \{A, N\}$, where A and N denote agriculture and nonagriculture, respectively. Households are dynastic and workers live for a single period: the worker inhabiting household i in period $t - 1$ is the parent of the worker inhabiting household i in period t . As such, workers are indexed by their parent's sector of employment in the previous period, $k \in \{A, N\}$. Accordingly, refer to workers whose parents worked in agriculture as ag-born and those whose parents worked in nonagriculture as nonag-born.

Heterogeneous Sectoral Skills Workers draw i.i.d. skills, $\epsilon_{i,j,t}$, for each sector $j \in \{A, N\}$ from the stationary Fréchet distribution:

$$\epsilon_{i,j,t} \sim F(\epsilon) \equiv \exp[-\epsilon^{-\gamma}]$$

⁹The utility shifters capture the idea that workers may have incentives for pursuing certain sectors beyond just the wage they can earn. For example, [Caselli and Coleman II \(2001\)](#) (and various papers thereafter including [Alonso-Carrera and Raurich \(2018\)](#); [Porzio, Rossi and Santangelo \(2022\)](#)) consider costs of pursuing nonagricultural work that workers incur but firms do not.

¹⁰Related evidence attributes occupational persistence to the transmission of occupation-specific human capital, access to employers and professions, information, and exposure ([Laband and Lentz, 1983](#); [Corak and Piraino, 2011](#); [Aina and Nicoletti, 2018](#); [Bell et al., 2019](#); [Haeck and Laliberté, 2025](#)).

The shape parameter $\gamma > 1$ governs the inverse of skill dispersion: as $\gamma \rightarrow 1$ skills become more dispersed and as $\gamma \rightarrow \infty$ the skill distribution collapses to a single point.

Inherited Productivity Wedges Each worker inherits a productivity wedge, $\kappa_j \geq 1$, which rescales their skill for their parent's sector. Then, the talent provided by each worker to a given sector is given by:

$$h_{i,j,t} = \epsilon_{i,j,t} \cdot \begin{cases} \kappa_j & \text{if } s_{i,t-1} = j \\ 1 & \text{otherwise} \end{cases}$$

Sectoral Amenities and the Inherited Immobility Wedge Working in sector j provides a common sectoral payoff shifter, $B_{j,t} > 0$, which scales workers' indirect utility without affecting their productive talent or labor income. Each ag-born worker also inherits an immobility wedge, $\tau \in [0, 1)$, which reduces their indirect utility from working in nonagriculture relative to nonag-born workers of equal skill. Let $V_{i,j,t}$ denote the indirect utility of worker i from working in sector j in period t . Then, the indirect utility of working in sector j is given by:

$$V_{i,j,t} = B_{j,t} \cdot w_{j,t} \cdot h_{i,j,t} \cdot \begin{cases} (1 - \tau) & \text{if } s_{i,t-1} = A \text{ and } j = N \\ 1 & \text{otherwise} \end{cases}$$

where $y_{i,j,t} \equiv w_{j,t} \cdot h_{i,j,t}$ is the labor income of worker i in sector j and $w_{j,t}$ is the wage per efficiency unit in sector j that workers take as given.

Worker's Problem Workers choose their sector to maximize indirect utility:

$$s_{i,t} = \arg \max_{j \in \{A, N\}} V_{i,j,t}$$

2.2. Optimization

Workers choose to work in agriculture if and only if:

$$\frac{\epsilon_{i,A,t}}{\epsilon_{i,N,t}} \cdot \begin{cases} \kappa_A \\ \frac{1}{\kappa_N} \end{cases} \geq \frac{B_{N,t}}{B_{A,t}} \cdot \frac{w_{N,t}}{w_{A,t}} \cdot \begin{cases} (1 - \tau) & \text{if } s_{i,t-1} = A \\ 1 & \text{if } s_{i,t-1} = N \end{cases}$$

The left-hand-side of the inequality represents workers' comparative advantage, while the right-hand-side represents the indirect utility threshold around which they sort. All else equal, the inherited agricultural productivity wedge κ_A and immobility wedge τ increase ag-born workers' incentives to choose agriculture. Likewise, the inherited nonagricultural productivity wedge κ_N raises nonag-born workers' incentives to choose nonagriculture.

Sector Allocations of Ag-born versus Nonag-born Workers The fractions of ag-born and nonag-born workers who choose to work in agriculture are given by:

$$\begin{aligned} \phi_{A,t}(k) &\equiv \mathbb{P}_i[s_{i,t} = A \mid s_{i,t-1} = k] \\ &= \begin{cases} \frac{1}{1 + \left[\frac{B_{N,t}}{B_{A,t}} \cdot \frac{w_{N,t}}{w_{A,t}} \cdot \frac{(1-\tau)}{\kappa_A} \right]^\gamma} & \text{if } k = A \\ \frac{1}{1 + \left[\frac{B_{N,t}}{B_{A,t}} \cdot \frac{w_{N,t}}{w_{A,t}} \cdot \kappa_N \right]^\gamma} & k = N \end{cases} \end{aligned} \quad (1)$$

where $k \in \{A, N\}$ indexes the worker's parent sector. Expressions analogous to (1) hold for the fractions of ag-born and nonag-born workers in nonagriculture, which also satisfy:

$$\phi_{N,t}(k) = 1 - \phi_{A,t}(k)$$

I visualize how inherited productivity and immobility wedges influence worker sorting with the numerical examples in Figure 2.

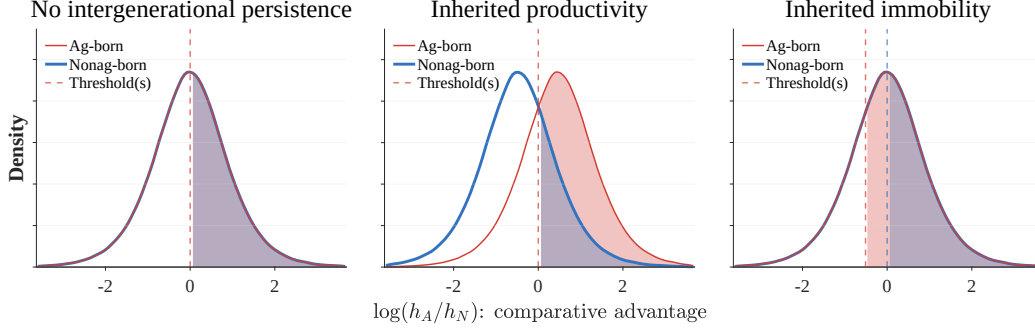


Figure 2: How Does Intergenerational Persistence Influence Sorting?

Notes: The red and blue curves depict the simulated distributions of log comparative advantage for ag-born and nonag-born workers, respectively, with each distribution normalized to integrate to one. Shading identifies the workers to the right of their group-specific dashed threshold who choose agriculture. The first panel sets $\kappa_A = \kappa_N = 1$ and $\tau = 0$, the second introduces inherited productivity through $\kappa_A, \kappa_N > 1$, and the third introduces inherited immobility through $\tau > 0$ while holding $\kappa_A = \kappa_N = 1$. Wages and amenities are held fixed across panels, and inherited immobility changes only the threshold faced by ag-born workers.

In this example, equally-sized ag-born and nonag-born subcohorts face relative wages and payoff shifters that, in the absence of intergenerational persistence, would lead to half of each subcohort choosing agriculture. This scenario is depicted in the first panel of Figure 2: the sorting of ag-born (red) and nonag-born (blue) workers into agriculture (denoted by the shaded area) is identical. In the second panel, I introduce inherited productivity through $\kappa_A, \kappa_N > 1$, which shifts the comparative advantage distribution of ag-born workers toward agriculture and that of nonag-born workers away from agriculture. Hence, the fraction of ag-born workers who choose agriculture increases, while the fraction of nonag-born workers who choose agriculture decreases. In the third panel, I introduce inherited immobility through $\tau > 0$ while holding $\kappa_A = \kappa_N = 1$. This shifts the threshold faced exclusively by ag-born workers toward lower comparative advantage in agriculture, which increases the fraction of ag-born workers who choose agriculture without affecting the fraction of nonag-born workers who choose agriculture.

Effective Labor Supplied by Ag-born and Nonag-born Workers The average effective labor supplied by k -born workers to sector j is given by:

$$h_{j,t}(k) \equiv \mathbb{E}_i [h_{i,j,t} \mid s_{i,t} = j, s_{i,t-1} = k] = \begin{cases} \kappa_j \cdot \bar{\gamma} \cdot \phi_{j,t}(k)^{-\frac{1}{\gamma}} & \text{if } j = k \\ \bar{\gamma} \cdot \phi_{j,t}(k)^{-\frac{1}{\gamma}} & j \neq k \end{cases} \quad (2)$$

where $\bar{\gamma} \equiv \Gamma\left(1 - \frac{1}{\gamma}\right)$ is a constant determined by the Gamma function $\Gamma(\cdot)$.

Two features of workers' average effective labor supply are worth highlighting. First, as in [Lagakos and Waugh \(2013\)](#), there are productivity gains from self-selection that are decreasing in the fraction of workers who sort into the sector.¹¹ Second, inherited productivity wedges *directly* increase the average effective labor supplied by k -born workers. Consider an economy with just the inherited productivity wedges $\kappa_A > 1$ and $\kappa_N > 1$. Figure 2 above illustrates that these wedges imply ag-born workers are more likely to choose agriculture than nonag-born workers. The direct influence of the inherited productivity wedge will raise the average skill (and moreover the average earnings) of ag-born workers in agriculture relative to nonag-born workers. However, nonag-born workers will benefit from higher gains from self-selection into agriculture than ag-born workers. As I outline below, I can identify the inherited productivity wedges from the average earnings gap between ag-born and nonag-born workers in each sector, but first I need to adjust for the self-selection effect.

2.3. Identification

There are four labor supply parameters to identify: the Fréchet shape parameter $\gamma > 1$, the inherited productivity wedges $\kappa_A \geq 1$ and $\kappa_N \geq 1$, and the inherited immobility wedge $\tau \in [0, 1)$. I exploit the structure of the model to identify these parameters from four moments concerning workers' income and sectoral allocations.

Fréchet Shape Parameter, $\gamma > 1$ Define the maximum indirect utility of a k -born worker i in their optimally chosen sector as $V_{i,t}^{\max}(k) \equiv \max_{j \in \{A,N\}} V_{i,j,t}$. This object is also Fréchet distributed with shape parameter γ , but with scale parameter $\Lambda_t(k)$,

¹¹To illustrate this, compare the average effective labor supplied by nonag-born workers to agriculture when they sort relative to an alternative environment where they are randomly assigned to sectors:

$$\frac{h_{A,t}(N)}{\mathbb{E}_i[\epsilon_{i,A,t} \mid s_{i,t-1} = N]} = \phi_{A,t}(N)^{-\frac{1}{\gamma}} > 1$$

Workers who sort into a sector are more productive than randomly assigned workers, but these gains from selection decline as more workers sort into the sector.

which is given by:

$$\Lambda_t(k) \equiv \begin{cases} \left[(B_{A,t}\kappa_A w_{A,t})^\gamma + (B_{N,t}(1-\tau)w_{N,t})^\gamma \right]^{\frac{1}{\gamma}} & \text{if } k = A \\ \left[(B_{A,t}w_{A,t})^\gamma + (B_{N,t}\kappa_N w_{N,t})^\gamma \right]^{\frac{1}{\gamma}} & k = N \end{cases}$$

A generic property of Fréchet distributed random variables is that they can be rewritten in terms of a standard Gumbel distributed random variable:

$$\log V_{i,t}^{\max}(k) = \log \Lambda_t(k) + \frac{1}{\gamma} u_{i,t}(k)$$

where $u_{i,t}(k)$ is Gumbel distributed with location parameter 0 and scale parameter 1. As shown in Appendix A.1.1, the maximum indirect utility is independent of the identity of the maximizing sector, so this distribution is unchanged after conditioning on sector choice.

Recall, each worker's income is equal to the skill they supply to their chosen sector multiplied by the wage per efficiency unit in that sector, $y_{i,j,t} \equiv w_{j,t} \cdot h_{i,j,t}$. Then, each k -born worker who optimally chooses sector j has realized labor income given by:

$$y_{i,j,t}(k) \equiv \begin{cases} \frac{V_{i,t}^{\max}(k)}{B_{j,t}(1-\tau)} & \text{if } j = N, k = A \\ \frac{V_{i,t}^{\max}(k)}{B_{j,t}} & \text{otherwise} \end{cases}$$

As such, logged individual income satisfies:

$$\log y_{i,j,t}(k) = \log \Lambda_t(k) - \log B_{j,t} - \mathbb{1}_i \{s_{i,t} = N, s_{i,t-1} = A\} \cdot \log(1-\tau) + \frac{1}{\gamma} u_{i,t}(k)$$

Since the variance of a standard Gumbel distributed random variable is $\pi^2/6$, then the Fréchet skill dispersion parameter γ is identified from the variance of logged individual income, conditional on sectoral choice and parental sector:

$$\text{Var} [\log y_{i,j,t}(k) \mid s_{i,t} = j, s_{i,t-1} = k] = \frac{\pi^2}{6} \cdot \frac{1}{\gamma^2} \quad (\text{ID-1})$$

Inherited Productivity Wedges, $\kappa_A \geq 1$ and $\kappa_N \geq 1$ Since each workers' labor income satisfies $y_{i,j,t} = w_{j,t} \cdot h_{i,j,t}$, the average labor income of k -born workers who

choose sector j is proportional to their average effective labor supply:

$$\begin{aligned} y_{j,t}(k) &\equiv \mathbb{E}_i [y_{i,j,t}(k) \mid s_{i,t} = j, s_{i,t-1} = k] \\ &= w_{j,t} \cdot \begin{cases} \kappa_j \cdot \bar{\gamma} \cdot \phi_{j,t}(k)^{-\frac{1}{\gamma}} & \text{if } k = j \\ \bar{\gamma} \cdot \phi_{j,t}(k)^{-\frac{1}{\gamma}} & k \neq j \end{cases} \end{aligned}$$

Thus, conditional on the inferred Fréchet shape parameter γ , the inherited productivity wedges κ_A and κ_N are identified from the selection-adjusted average income gap between ag-born and nonag-born workers in each sector:

$$\frac{y_{A,t}(A)}{y_{A,t}(N)} = \kappa_A \cdot \left(\frac{\phi_{A,t}(A)}{\phi_{A,t}(N)} \right)^{-\frac{1}{\gamma}} \quad (\text{ID-2})$$

and

$$\frac{y_{N,t}(A)}{y_{N,t}(N)} = \frac{1}{\kappa_N} \cdot \left(\frac{\phi_{N,t}(A)}{\phi_{N,t}(N)} \right)^{-\frac{1}{\gamma}} \quad (\text{ID-3})$$

Inherited Immobility Wedge, $\tau \in [0, 1)$ One measure of an economy's intergenerational persistence in sector choice is the odds ratio, which represents the relative likelihood that an ag-born worker chooses to work in agriculture compared to a nonag-born worker:

$$\mathcal{O} \equiv \frac{\phi_{A,t}(A)/\phi_{N,t}(A)}{\phi_{A,t}(N)/\phi_{N,t}(N)} = \frac{\left[\frac{B_{N,t}}{B_{A,t}} \cdot \frac{w_{N,t}}{w_{A,t}} \cdot \kappa_N \right]^\gamma}{\left[\frac{B_{N,t}}{B_{A,t}} \cdot \frac{w_{N,t}}{w_{A,t}} \cdot \frac{(1-\tau)}{\kappa_A} \right]^\gamma}$$

Then, conditional on the inferred Fréchet shape parameter γ and inherited productivity wedges κ_A and κ_N , the inherited immobility wedge τ is identified as the economy's only remaining source of persistence:

$$\mathcal{O} = \left(\frac{\kappa_A \cdot \kappa_N}{1 - \tau} \right)^\gamma \quad (\text{ID-4})$$

2.4. Estimation

To estimate these parameters, I use the linked full count historical census microdata from IPUMS ([Ruggles et al., 2024](#); [Helgertz et al., 2024](#)) with sample construction as described in section 1. Income data are only available in the 1940 census;

hence, I use only the linked sample from that year to estimate the labor supply parameters. I restrict the linked sample to individuals with positive wage and salary income, positive weeks worked, and at least 35 hours per week worked. In the appendix, I report estimates using annual or weekly earnings, which are broadly similar to my baseline estimates using hourly earnings.

There are two noteworthy limitations of these data. First, the census topcodes income at \$5,000 in 1940. Topcoded workers constitute 0.104 percent of the estimation sample, and their capped earnings constitute 0.501 percent of its total wage and salary income. Topcoding compresses measured log-income dispersion and may therefore lead to slight overestimates of the Fréchet shape parameter γ . Moreover, this would also lead to slight underestimates of the inherited productivity wedges κ_A and κ_N because I would attribute too much of the average income gaps to the relative gains from self-selection. As a robustness check, I multiply every topcoded observation by 2.926 before constructing hourly earnings. This factor makes the affected observations account for 1.452 percent of total earnings, matching the 1940 salary income share of the top 0.1 percent reported by [Atkinson, Piketty and Saez \(2011\)](#).

Second, with only wage and salary income reported in the census, I do not observe self-employed farming or business income. As such, in the linked sample I only observe income for about 46 percent of agricultural workers and 90 percent of nonagricultural workers. This is a serious limitation. Suppose, for example, that the most productive ag-born workers are more likely to sort into self-employed farming rather than wage work while nonag-born workers have no selection into self-employment. As such, the variance of observed earnings will be smaller than the population variance of earnings, which would lead to an upward bias in the estimate of γ . In addition, the inherited productivity wedge in agriculture, κ_A , would be underestimated primarily because the observed ag-born wage workers are less productive on average than the unobserved self-employed ag-born workers.¹² Moreover, this would result in an overestimate of the inherited immobility wedge τ . To proceed, I maintain the assumption that any selection into self-employment rather than wage work does not depend on latent productivity within

¹²There are two other potential sources of downward bias in the estimates of κ_A . First, excluding self-employed ag-born farmers from the sample may lead to an underestimate of the fraction of ag-born workers who choose agriculture, $\phi_{A,t}(A)$, which would bias κ_A downward. Second, the upward bias in the estimate of γ would also bias κ_A downward. Even without any selection into self-employment for nonag-born workers, these two sources of bias would also lead to an underestimate of the inherited productivity wedge in nonagriculture, κ_N .

sector-by-parental-sector cells.¹³ Henceforth, when using phrases like “selection-adjusted”, this refers to selection across sectors, not selection into self-employment versus wage work.

The observed income data may reflect other sources of heterogeneity that the model abstracts from including schooling, age, and local labor market conditions. My baseline parameter estimates are based on the identification strategy consistent with the model as outlined above; however, I report robustness checks that condition on these additional sources of heterogeneity. The results are broadly similar to the baseline estimates.

Estimates of γ As outlined above, the Fréchet shape parameter γ is identified from the variance of logged individual income, conditional on sectoral choice and parental sector. I report my preferred hourly estimate of γ in Table 3, together with estimates that add controls or adjust topcoded earnings. The baseline variance-based estimate is 2.076, while adding adult commuting-zone, childhood-schooling, and age fixed effects raises the estimate to 2.293. Rescaling topcoded earnings yields 2.063, very close to the baseline estimate. Appendix Table 12 reports corresponding estimates using annual and weekly earnings. These estimates are within the range of 1.5 to 2.5 typically found in the existing literature (e.g., [Kuralbayeva and Stefanski \(2013\)](#); [Hsieh et al. \(2019\)](#); [Eckert and Peters \(2022\)](#); [Alvarez \(2017\)](#)).

Table 3: Estimates of the Fréchet Shape Parameter: Hourly Earnings

	Baseline	Added FE	Topcode adjusted
$\hat{\gamma}$	2.076	2.293	2.063
Observations	2,077,253	2,077,253	2,077,253
Quantile RMSE (1st–99th percentiles)	0.337	0.307	0.340

Note: Columns 1 and 2 infer γ from the residual variance of log hourly earnings; column 2 additionally controls for adult commuting zone, childhood schooling, and age. Column 3 replaces each topcoded income with 2.926 times the censoring threshold, calibrated to match a 1.452 percent top-0.1-percent income share. All columns use 2,077,253 observations with positive earnings and weeks and at least 35 weekly hours; no log-earnings restriction is imposed. Quantile RMSE measures fit across the 1st–99th residual percentiles in log points; lower values indicate better fit.

For a visual check, Figure 3 compares the residual earnings distribution with

¹³[Carruthers and Wanamaker \(2017\)](#) discuss the analogous limitation created by the absence of self-employment income in the 1940 Census and likewise exclude individuals without recorded earnings from their baseline analysis.

the log-Fréchet densities implied by the baseline and topcode-adjusted estimates. Appendix A.1 reports results for alternative earnings measures, sensitivity to the topcode adjustment, and a censored-likelihood estimator.

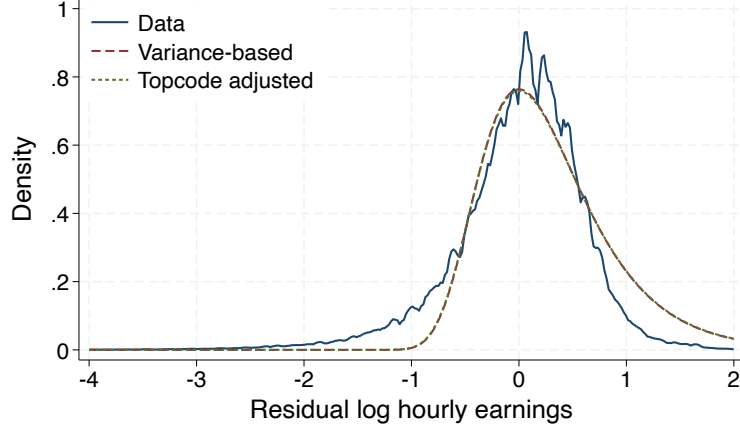


Figure 3: Residual Earnings Distribution in Data and Model

Notes: The solid line plots the kernel density of residual log hourly wage and salary income after removing worker-sector-by-parental-sector cell means. The dashed red and short-dashed green lines use the baseline and topcode-adjusted estimates, $\hat{\gamma} = 2.076$ and 2.063, respectively. The estimation sample has positive wage and salary income, positive weeks worked, and at least 35 usual weekly hours, with no log-earnings restriction; the figure displays residuals between -4 and 2.

Estimates of Inherited Productivity and Immobility Wedges Next, I estimate the inherited productivity wedges κ_A and κ_N from selection-adjusted average income differences between ag-born and nonag-born workers employed in the same sector. Then, I estimate the inherited immobility wedge τ from the remaining difference between the sector choices of ag-born and nonag-born workers. I report estimates using annual and weekly income in Appendix Table 13.

Let $\hat{y}_j(k)$ denote mean income in the relevant income sample among workers employed in sector j whose fathers worked in sector k , and let $\hat{\phi}_j(k)$ denote the corresponding sector-choice probability in the full linked sample. Given the estimated $\hat{\gamma}$, I compute the inherited productivity wedges as:

$$\hat{\kappa}_A = \frac{\hat{y}_A(A)/\hat{\phi}_A(A)^{-\frac{1}{\hat{\gamma}}}}{\hat{y}_A(N)/\hat{\phi}_A(N)^{-\frac{1}{\hat{\gamma}}}} \quad \text{and} \quad \hat{\kappa}_N = \frac{\hat{y}_N(N)/\hat{\phi}_N(N)^{-\frac{1}{\hat{\gamma}}}}{\hat{y}_N(A)/\hat{\phi}_N(A)^{-\frac{1}{\hat{\gamma}}}}$$

Hence, the inherited productivity is identified by the ratio of average income between ag-born and nonag-born workers in the same sector after removing the gains from self-selection.

For example, the average hourly earnings of ag-born workers in agriculture is 0.16 and for nonag-born workers in agriculture is 0.34. The gains from self-selection play a critical role here. For ag-born workers, their average income in agriculture is roughly 35% higher due to gains from self-selection. However, for nonag-born workers, their average income in agriculture is roughly 312% higher due to gains from self-selection. After adjusting for these selection effects, the average income of ag-born workers in agriculture is roughly 42% higher than that of nonag-born workers in agriculture, which is due to the inherited productivity wedge, $\hat{\kappa}_A$. Similarly, the selection-adjusted average income of nonag-born workers in nonagriculture is roughly 81% higher than that of ag-born workers in nonagriculture, which is due to the inherited productivity wedge, $\hat{\kappa}_N$.

Define the sample sector-choice odds ratio as:

$$\hat{\mathcal{O}} \equiv \frac{\hat{\phi}_A(A)/\hat{\phi}_N(A)}{\hat{\phi}_A(N)/\hat{\phi}_N(N)}.$$

Then, the inherited immobility wedge is identified as the remaining source of inter-generational persistence in sector choice not explained by inherited productivity wedges:

$$\hat{\tau} = 1 - \frac{\hat{\kappa}_A \hat{\kappa}_N}{\hat{\mathcal{O}}^{1/\hat{\gamma}}}.$$

Using sector choices from the full linked sample throughout the wedge inversion, I find that ag-born workers sort as if they value nonagricultural work about 40 percent less than an equally-skilled nonag-born worker. Expressed as shares of observed log-odds persistence, inherited productivity accounts for 64.8 percent and inherited immobility for 35.2 percent; Appendix A.1.2 derives this decomposition. Using the topcode-adjusted estimate of γ yields nearly identical estimates: $\hat{\kappa}_A = 1.434$, $\hat{\kappa}_N = 1.809$, and $\hat{\tau} = 0.401$. I report the estimates of κ_A , κ_N , and τ in Table 4.

Table 4: Inherited Productivity and Immobility Wedges: Hourly Earnings

	Baseline	Added FE	Topcode adjusted
<i>Panel A: Inferred wedges</i>			
Inherited ag. productivity, $\hat{\kappa}_A$	1.424	1.205	1.434
Inherited nonag. productivity, $\hat{\kappa}_N$	1.805	1.431	1.809
Inherited mobility wedge, $\hat{\tau}$	0.401	0.341	0.401
<i>Panel B: Mean hourly earnings</i>			
Agriculture, ag-born: $\hat{y}_A(A)$	0.16	0.17	0.16
Agriculture, nonag-born: $\hat{y}_A(N)$	0.34	0.29	0.34
Nonagriculture, ag-born: $\hat{y}_N(A)$	0.46	0.51	0.46
Nonagriculture, nonag-born: $\hat{y}_N(N)$	0.60	0.58	0.60
<i>Panel C: Intergenerational persistence</i>			
Ag. employment share, ag-born	0.533	0.471	0.533
Ag. employment share, nonag-born	0.053	0.089	0.053
Sector-choice odds ratio	20.541	9.095	20.541

Note: The table reports inherited productivity premia, $\hat{\kappa}_A$ and $\hat{\kappa}_N$, and the inherited immobility wedge, $\hat{\tau}$, using hourly earnings from the 1940 linked census. Columns 1–3 use the baseline, added-fixed-effect, and topcode-adjusted estimates of γ from Table 3; column 2 also adjusts the underlying moments for adult commuting zone, childhood schooling, and age. Earnings moments entering κ_j use the income sample defined in Table 3; all sector-choice moments use the full sample of workers with valid worker and parental sectors. Panel B reports the income-sample means, while Panel C reports the full-sample sector choices also used to estimate κ_j and τ . Column 3 holds the baseline moments fixed and changes only $\hat{\gamma}$.

Discussion Although the linked census data alone could not distinguish the underlying sources of intergenerational persistence, the model structure allows me to separately identify inherited productivity and immobility wedges. My estimates suggest that ag-born workers are disproportionately drawn to agriculture both because they are more productive in agriculture and because they find nonagricultural work less appealing than nonag-born workers. Through the lens of my model, my baseline estimates suggest that ag-born workers are about 42 percent more productive in agriculture than nonag-born workers, while nonag-born workers are about 81 percent more productive in nonagriculture than ag-born workers. In addition, ag-born workers value nonagricultural work about 40 percent less than an equally-skilled nonag-born worker. The next section embeds this labor-supply block in general equilibrium, and the subsequent quantitative analysis uses

the resulting model to measure how inherited productivity and immobility shaped U.S. structural transformation.

3. General Equilibrium Closure

Here I describe the general equilibrium closure of the model including the aggregation of the labor supply block and the production side of the economy. Then, I define the competitive equilibrium and characterize how intergenerational persistence alters the economy's response to the drivers of structural transformation.

3.1. Aggregation

Equations (1) and (2) in section 2 describe the allocations of workers and effective labor among ag-born and nonag-born workers, respectively. However, the aggregate allocations of workers and effective labor across sectors are the relevant objects for structural transformation. Notably, I continue to abstract from any incumbent worker reallocation: at all times the economy comprises a single generation of workers.

Worker Allocations The fraction of workers in agriculture is given by:

$$\phi_{A,t} = \phi_{A,t-1} \cdot \phi_{A,t}(A) + (1 - \phi_{A,t-1}) \cdot \phi_{A,t}(N) \quad (3)$$

There is an analogous expression for the fraction of workers in nonagriculture, which moreover satisfies: $\phi_{N,t} = 1 - \phi_{A,t}$. The aggregate worker allocation depends both on the choices of ag-born and nonag-born workers as well as the composition of the cohort, which is determined by the aggregate worker allocation in the previous period. Hence, even though workers solve a static sector choice problem, the evolution of the aggregate worker allocation is dynamic. As I will discuss in greater detail below, this feature of the economy is a key channel through which intergenerational persistence uniquely shapes the dynamics of structural transformation.

Effective Labor Supply The total effective labor supply in the agricultural sector is given by:

$$H_{A,t} = \phi_{A,t-1} \cdot \phi_{A,t}(A) \cdot h_{A,t}(A) + (1 - \phi_{A,t-1}) \cdot \phi_{A,t}(N) \cdot h_{A,t}(N) \quad (4)$$

There is an analogous expression for the total effective labor supply in nonagriculture, $H_{N,t}$. The composition of ag-born versus nonag-born workers also affects the aggregate effective labor supply in each sector. Moreover, the effective labor supply in each sector depends directly on the inherited productivity wedges. As such, past worker reallocation will transmit indirectly to future generations through the composition of ag-born and nonag-born workers via the aggregate effective labor supply, and this will depend on the inherited productivity wedges. I will formalize this when characterizing how intergenerational persistence shapes the dynamics of structural transformation in general equilibrium.

Aggregate Income The total labor income in the economy is given by:

$$Y_t = w_{A,t} \cdot H_{A,t} + w_{N,t} \cdot H_{N,t} \quad (5)$$

3.2. Production

The economy features representative firms in each sector as well as a final good producer that aggregates the sectoral goods into a final consumption bundle. I assume that the final consumption bundle is distributed to workers according to their share of aggregate income.

Sectoral Production Each sector has a representative firm that produces output using labor as the only factor input according to:

$$Q_{A,t} = Z_{A,t} \cdot H_{A,t} \quad \text{and} \quad Q_{N,t} = Z_{N,t} \cdot H_{N,t} \quad (6)$$

where $Z_{j,t} > 0$ is the production technology of sector j .

Labor Demand The firm takes the price of goods, $p_{j,t}$, and the wage per efficiency unit of labor, $w_{j,t}$, as given, and chooses labor demand to maximize profits:

$$\max_{H_{j,t}} p_{j,t} Q_{j,t} - w_{j,t} H_{j,t}$$

which implies that the wage per efficiency unit of labor is equal to the value of the marginal product of labor:

$$w_{A,t} = p_{A,t} \cdot Z_{A,t} \quad \text{and} \quad w_{N,t} = p_{N,t} \cdot Z_{N,t} \quad (7)$$

Final Good Producer There is a final good producer that demands the sectoral goods, $X_{j,t}$, and aggregates them into a final consumption bundle, Q_t , according to the nonhomothetic CES production function described in [Comin, Lashkari and Mestieri \(2021\)](#), which extends the demand-based structural-transformation mechanisms in [Kongsamut, Rebelo and Xie \(2001\)](#) and [Boppart \(2014\)](#):

$$\sum_{j \in \{A, N\}} \Omega_j^{1/\sigma} \left(\frac{X_{j,t}}{Q_t^{\vartheta_j}} \right)^{\frac{\sigma-1}{\sigma}} = 1, \quad (8)$$

where $\Omega_j > 0$ is the demand weight on sector j , $\sigma \in (0, 1)$, and $\vartheta_N > \vartheta_A = 1$. If $\vartheta_A = \vartheta_N = 1$, then this recovers the typical homothetic CES aggregator with elasticity of substitution σ .

Define the sector j output share as $\omega_{j,t} \equiv p_{j,t}X_{j,t}/(P_tQ_t)$, where P_t is the price of the final consumption bundle satisfying:

$$\frac{Y_t}{P_t} = Q_t \quad (9)$$

Then, relative quantity demand satisfies:

$$\frac{X_{A,t}}{X_{N,t}} = \frac{\Omega_A}{\Omega_N} \left(\frac{p_{N,t}}{p_{A,t}} \right)^{\sigma} \left(\frac{Y_t}{P_t} \right)^{(\vartheta_A - \vartheta_N)(1-\sigma)}. \quad (10)$$

Multiplying relative quantity demand by $p_{A,t}/p_{N,t}$ gives relative output shares:

$$\frac{\omega_{A,t}}{\omega_{N,t}} = \frac{\Omega_A}{\Omega_N} \left(\frac{p_{N,t}}{p_{A,t}} \right)^{\sigma-1} \left(\frac{Y_t}{P_t} \right)^{(\vartheta_A - \vartheta_N)(1-\sigma)}. \quad (11)$$

3.3. Competitive Equilibrium

Let $p_{A,t} = 1$ be the numeraire and $p_t \equiv p_{N,t}/p_{A,t}$ be the relative price of nonagricultural goods. For an initial worker allocation $\phi_{A,0}$ and the exogenous fundamentals

$$\{Z_{j,t}, B_{j,t}\}_{j \in \{A, N\}, t > 0}$$

a competitive equilibrium is a sequence of allocations

$$\{\phi_{j,t}(k), h_{j,t}(k), \phi_{j,t}, H_{j,t}, Y_t, Q_t, X_{j,t}, \omega_{j,t}, Q_t\}_{j,k \in \{A, N\}, t > 0}$$

and prices

$$\{w_{j,t}, p_t, P_t\}_{j \in \{A, N\}, t > 0}$$

such that:

1. (Non)ag-born workers optimally choose sectors according to equation (1) such that the aggregate allocation satisfies equation (3).
2. The average effective labor supplied to each sector by ag-born and nonag-born workers satisfies equation (2), and the total effective labor supplied to each sector satisfies equation (4).
3. Aggregate labor income satisfies equation (5).
4. Competitive sectoral firms produce according to equation (6) and pay each efficiency unit its value marginal product according to equation (7).
5. The final good producer chooses sectoral inputs according to equation (11) and (10) such that final good output satisfies equation (8), and the final-good price index satisfies equation (9).
6. Sectoral goods markets clear:

$$X_{j,t} = Q_{j,t} \quad \text{for each} \quad j \in \{A, N\}. \quad (12)$$

given the parameters $\{\gamma, \kappa_A, \kappa_N, \tau, \sigma, \vartheta_A, \vartheta_N, \Omega_A, \Omega_N\}$.

3.4. Characterizing the Influence of Intergenerational Persistence

In this model, all worker reallocation results from either a labor demand shock originating in sectoral technologies (i.e., $Z_{j,t}$) or a labor supply shock originating in the payoff shifter (i.e., $B_{j,t}$). For expositional simplicity, I focus on a 1-percent permanent improvement in the agricultural technology. To explore how intergenerational persistence shapes the economy's response to this shock, I simplify the transmission into three components depicted in the figure below.

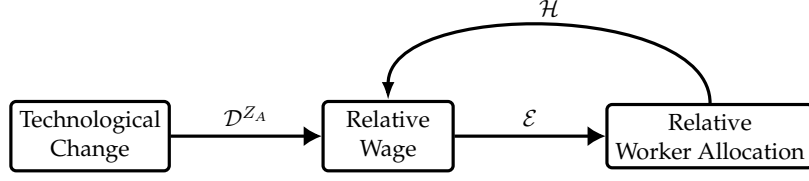


Figure 4: Diagram of the Contemporaneous Transmission of Technological Change

Notes: $\mathcal{D}^{Z_A}$ is the technology-induced shift in relative labor demand, $\mathcal{E}$ is the elasticity of relative worker allocation to the relative wage, and $\mathcal{H}$ is the relative-wage feedback from worker reallocation.

To characterize the contemporaneous equilibrium response, I reduce the equilibrium system to two local equations: a relative headcount labor-supply schedule and an inverse relative labor-demand schedule expressed in terms of headcount allocations. The relative labor-supply schedule is

$$d \log \frac{\phi_{N,t}}{\phi_{A,t}} = \mathcal{E} d \log \frac{w_{N,t}}{w_{A,t}},$$

where $\mathcal{E}$ measures the elasticity of workers' relative sectoral allocation with respect to the relative wage, holding cohort composition and the relative payoff shifter $B_{N,t}/B_{A,t}$ fixed. The inverse relative labor-demand schedule is

$$d \log \frac{w_{N,t}}{w_{A,t}} = \mathcal{D}^{Z_A} d \log Z_{A,t} + \mathcal{H} d \log \frac{\phi_{N,t}}{\phi_{A,t}}.$$

Here $\mathcal{D}^{Z_A}$ measures the shift in relative labor demand generated by agricultural technological change, holding sectoral effective labor supplies fixed. The slope $\mathcal{H}$ captures how worker reallocation feeds back into the relative wage through changes in sectoral effective labor supplies, production, and relative goods prices, holding technologies fixed. ¹⁴

Solving the two schedules gives the full general-equilibrium response of the relative worker allocation:

$$d \log \frac{\phi_{N,t}}{\phi_{A,t}} = \frac{\mathcal{E}}{1 - \mathcal{E}\mathcal{H}} \mathcal{D}^{Z_A} d \log Z_{A,t}.$$

Because $d\phi_{A,t} = -\phi_{A,t}(1 - \phi_{A,t}) d \log(\phi_{N,t}/\phi_{A,t})$, the contemporaneous reallocation

¹⁴Appendix A.2 formally defines these objects and derives both schedules from the full equilibrium system.

of agricultural workers is therefore

$$\frac{d\phi_{A,t}}{d \log Z_{A,t}} = -\phi_{A,t}(1 - \phi_{A,t}) \frac{\mathcal{E}}{1 - \mathcal{E}\mathcal{H}} \mathcal{D}^{Z_A}.$$

I use this decomposition to compare contemporaneous responses across economies with different sources of intergenerational persistence.

Stylized Economies Refer to the first economy as the “benchmark” economy, which has no intergenerational persistence (i.e., $\kappa_A = \kappa_N = 1$ and $\tau = 0$). Refer to the second economy as the “IGP- κ ” economy, which has intergenerational persistence from a symmetric inherited productivity only (i.e., $\kappa_A = \kappa_N = \kappa > 1$, and $\tau = 0$). Refer to the third economy as the “IGP- τ ” economy, which has intergenerational persistence from inherited immobility only (i.e., $\kappa_A = \kappa_N = 1$ and $\tau > 0$). I restrict the two IGP economies to have the same gap in the allocations of ag-born and nonag-born workers:

$$\phi_{A,t}(A)^{\text{IGP-}\kappa} - \phi_{A,t}(N)^{\text{IGP-}\kappa} = \phi_{A,t}(A)^{\text{IGP-}\tau} - \phi_{A,t}(N)^{\text{IGP-}\tau} > 0$$

Finally, I require that all three economies have the same stationary allocation of workers absent any technological change, $\phi_{A,t} = \phi_{A,t-1}$.

How Workers Respond to Wages in Partial Equilibrium, $\mathcal{E}$ The main channel through which intergenerational persistence shapes the economy’s immediate response to technological change is through the elasticity of worker allocation with respect to the relative wage, $\mathcal{E}$. In particular, this elasticity is given by:

$$\mathcal{E} = \gamma \cdot \left[1 - \left(\phi_{A,t}(A) - \phi_{A,t}(N) \right)^2 \right] > 0$$

Without any intergenerational persistence, the gap in sectoral allocations between ag-born and nonag-born workers is zero, and this elasticity is equal to γ . Otherwise, ag-born workers are more likely to work in agriculture than nonag-born workers, and the elasticity is strictly less than γ . However, that reduction in workers’ responsiveness to wages is independent of the source of intergenerational persistence. I formalize this in the following result.

Lemma 1. *The elasticity of worker allocation with respect to the relative wage is strictly*

smaller in the presence of intergenerational persistence regardless of whether that persistence arises from inherited productivity or inherited immobility.

$$0 < \mathcal{E}^{IGP-\kappa} = \mathcal{E}^{IGP-\tau} < \mathcal{E}^{bench} = \gamma$$

Recall, I documented a 67 pp. gap in the allocations of ag-born and nonag-born workers for the 1880 U.S. in section 1. This observed intergenerational persistence suggests workers are roughly 45% less responsive to changes in wages than they would be in the benchmark economy. When implementing the quantitative analysis, I will show that this reduction in workers' responsiveness to wages is an important way in which intergenerational persistence slows the pace of structural transformation.

How Workers Respond to Technical Change in General Equilibrium In order to provide a transparent characterization of how intergenerational persistence shapes workers' contemporaneous response to technological change, I restrict the production-side of the economy to be homothetic (i.e., $\vartheta_A = \vartheta_N = 1$). This shuts down the aggregate income channel through which changes in effective labor supply feed back into the relative wage, but preserves the relative price feedback channel. With this restriction, the general equilibrium feedback of worker reallocation into the relative wage is greater when the economy features intergenerational persistence, especially from inherited immobility. Formally:

Lemma 2. *The magnitude of the general equilibrium feedback of worker reallocation into the relative wage is strictly larger in the presence of intergenerational persistence, and it is largest when that persistence arises from inherited immobility.*

$$\mathcal{H}^{IGP-\tau} < \mathcal{H}^{IGP-\kappa} \leq \mathcal{H}^{bench} = -\frac{1}{\sigma} \cdot \frac{\gamma - 1}{\gamma}$$

where the equality holds only when $\phi_{A,t-1} = 0.5$.

Workers reallocating out of agriculture implies a reallocation of effective labor that induces a decrease in the relative wage. However, the magnitude of that general equilibrium feedback is larger in the presence of intergenerational persistence, especially due to inherited immobility. This together with the reduction in workers' responsiveness to wages, $\mathcal{E}$, implies that the contemporaneous response of workers to technological change is smaller in the presence of intergenerational persistence.

Proposition 1. *Improvements in agricultural technology induce a reallocation of workers out of agriculture, but the magnitude of that reallocation is smaller in the presence of intergenerational persistence.*

$$\frac{d\phi_{A,t}}{d\log Z_{A,t}}^{\text{bench}} < \frac{d\phi_{A,t}}{d\log Z_{A,t}}^{\text{IGP-}\kappa} < \frac{d\phi_{A,t}}{d\log Z_{A,t}}^{\text{IGP-}\tau} < 0$$

In particular, workers reallocate less when the economy features inherited immobility than when it features inherited productivity.

Figure 5 illustrates the equilibrium forces underlying Proposition 1. Because the relative headcount labor-supply schedule has slope $1/\mathcal{E}$, the smaller worker-allocation elasticity in the two IGP economies makes their supply schedules steeper than in the benchmark economy. The common allocation-gap restriction makes these schedules identical in the IGP- κ and IGP- τ economies. The source of persistence instead distinguishes their inverse labor-demand schedules: the more negative $\mathcal{H}$ in the IGP- τ economy produces a steeper downward-sloping schedule than in the IGP- κ economy, whose schedule is in turn steeper than in the benchmark economy. Consequently, the same technology-induced shift in relative labor demand generates the largest increase in $\log(\phi_N/\phi_A)$ in the benchmark economy, a smaller increase under inherited productivity, and the smallest increase under inherited immobility. Thus intergenerational persistence dampens contemporaneous structural transformation through both a weaker headcount-supply response and a stronger relative-wage feedback, with inherited immobility producing the greatest dampening because its feedback is strongest.

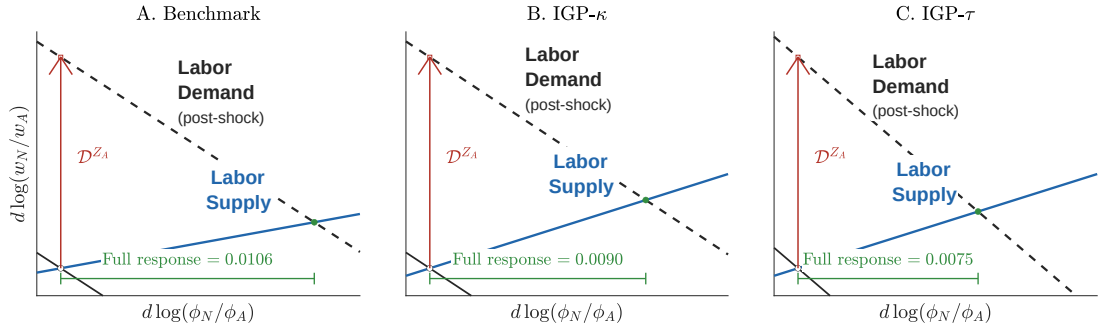


Figure 5: Headcount Relative Labor Supply and Demand with a Shock to Agricultural Technology

Notes: The figure plots log changes in the nonagricultural-to-agricultural employment ratio against log changes in the relative wage, normalizing the initial equilibrium to the origin. Blue lines show relative labor supply; solid and dashed black lines show inverse labor demand before and after a one-percent increase in agricultural technology. The red arrow is the demand shift, the green circle the new equilibrium, and the green bracket the resulting worker reallocation. All panels impose homothetic demand, $\gamma = 2.076$, $\sigma = 0.3$, and $\phi_A = 0.665$. Panel A has no intergenerational persistence; Panels B and C generate an odds ratio of 25 entirely through symmetric inherited productivity ($\kappa_A = \kappa_N = 2.171$) or inherited immobility ($\tau = 0.788$). These two economies have the same allocation gap but different general-equilibrium wage feedbacks.

Future Generations Respond to Past Reallocation Past worker reallocation alters the composition of ag-born and nonag-born workers in subsequent generations. In particular, if workers are induced to leave agriculture in the current generation, then there are fewer ag-born workers in the next generation. This composition change produces two initial shifts in the period- $t + 1$ equilibrium. It directly changes the aggregate headcount allocation by changing the weights placed on ag-born and nonag-born workers, and it indirectly changes the relative wage by changing the effective labor supplied by those groups. Define these two shifts as

$$\mathcal{F}^{direct} \equiv \left. \frac{\partial \phi_{A,t+1}}{\partial \phi_{A,t}} \right|_{d\phi_{j,t+1}(k)=0} \quad \text{and} \quad \mathcal{F}^{indirect} \equiv \left. \frac{\partial \log(w_{N,t+1}/w_{A,t+1})}{\partial \phi_{A,t}} \right|_{d\phi_{j,t+1}(k)=0}.$$

Combining these two shifts with the contemporaneous worker-allocation and relative-wage feedback characterized above gives the total dynamic response: ¹⁵

$$\frac{d\phi_{A,t+1}}{d\phi_{A,t}} = \mathcal{F}^{direct} - \phi_{A,t+1}\phi_{N,t+1} \cdot \frac{\mathcal{E}}{1 - \mathcal{E}\mathcal{H}} \cdot \mathcal{F}^{indirect}$$

First, the composition of ag-born and nonag-born workers directly affects the

¹⁵ Appendix A.2 derives this expression from the full period- $t + 1$ equilibrium system.

aggregate worker allocation through $\mathcal{F}^{direct}$. Ag-born workers are more likely to work in agriculture than nonag-born workers, so a smaller fraction of ag-born workers implies a smaller fraction of workers in agriculture. In particular, holding wages fixed, workers respond directly to past reallocation according to:

$$\mathcal{F}^{direct} = \phi_{A,t+1}(A) - \phi_{A,t+1}(N)$$

This direct transmission of past worker reallocation is nonzero only when the economy features intergenerational persistence, though it is independent of the source of that persistence. Formally:

Lemma 3. *Holding wages fixed, future generations respond to past reallocation according to:*

$$0 = \mathcal{F}^{direct,bench} < \mathcal{F}^{direct,IGP-\kappa} = \mathcal{F}^{direct,IGP-\tau}$$

Hence, intergenerational persistence directly propagates past reallocation forward to future generations.

Second, the composition of ag-born and nonag-born workers also affects the effective labor supplied to each sector through equation (4). As such, a reduction in ag-born workers implies a decrease (increase) in the effective labor supplied to (non)agriculture in the next generation, all else equal. This moreover induces a change in the relative wage, which leads to a reallocation of workers. By restricting the production side of the economy to be homothetic, I provide a transparent characterization of how intergenerational persistence shapes the indirect transmission of past worker reallocation to future generations through the relative wage.

Lemma 4. *Past worker reallocation induces an indirect worker-allocation response through the period- $t + 1$ relative wage when the economy features intergenerational persistence.*

$$\mathcal{F}^{indirect,IGP-\kappa} > \mathcal{F}^{indirect,IGP-\tau} > \mathcal{F}^{indirect,bench} = 0$$

The magnitude of this transmission is larger when the economy features inherited productivity than when it features inherited immobility.

With fewer ag-born workers due to a previous allocation of workers out of agriculture, the effective labor supply tilts toward nonagriculture, which leads to a reduction in the relative wage. For the IGP- τ economy, the relative-wage reduction offsets part, but not all, of the direct propagation of past worker reallocation. Thus

workers leaving agriculture in the current generation induce workers to leave agriculture in the next generation. In the IGP- κ economy, the relative-wage reduction is stronger and may or may not overcome the direct propagation channel. Consequently, the sign of the total response is ambiguous and depends on the relative magnitudes of the two channels.

Proposition 2. *Past worker reallocation alters the composition of ag-born and nonag-born workers in future generations, which in turn affects the aggregate worker allocation in those generations.*

$$\left(\frac{d\phi_{A,t+1}}{d\phi_{A,t}} \right)^{IGP-\tau} > 0, \quad \left(\frac{d\phi_{A,t+1}}{d\phi_{A,t}} \right)^{bench} = 0.$$

In the IGP- κ economy, the derivative can be positive or negative depending on whether the direct composition effect or the induced relative-wage effect dominates.

I illustrate the ambiguity of the dynamic transmission of past worker reallocation in the IGP- κ economy in figure 6 below. For most empirically relevant values of the inherited productivity wedge, the direct composition effect is dominated by the induced relative-wage effect, and past worker reallocation out of agriculture induces future generations to return to agriculture. For large enough κ values, the direct composition effect dominates, and past worker reallocation out of agriculture induces future generations to leave agriculture as well. In contrast, the IGP- τ economy always features a positive dynamic derivative, so past worker reallocation out of agriculture induces future generations to leave agriculture as well.

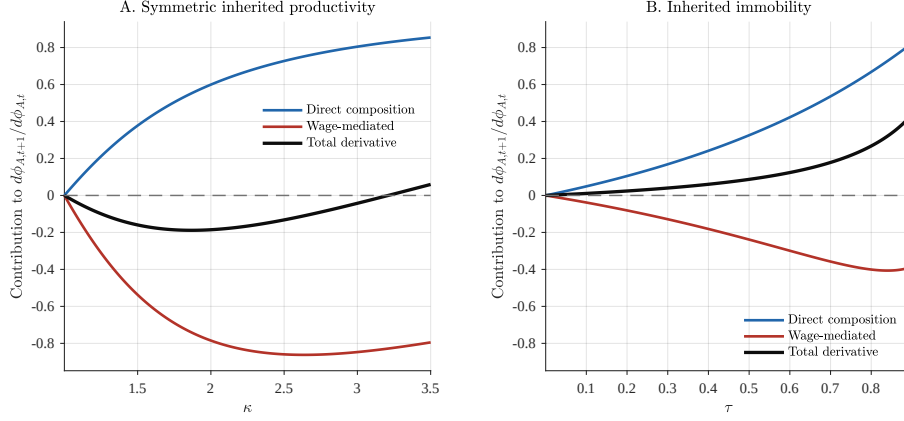


Figure 6: How Does Past Worker Reallocation Affect Future Generations?

Notes: The figure plots $d\phi_{A,t+1}/d\phi_{A,t}$ under homothetic demand, with $\phi_A = 0.665$, $\gamma = 2.076$, and $\sigma = 0.3$. Panel A varies symmetric inherited productivity, $\kappa_A = \kappa_N = \kappa \in (1, 3.5]$, with $\tau = 0$; Panel B varies inherited immobility, $\tau \in (0, 0.9]$, with $\kappa_A = \kappa_N = 1$. At each value, I solve for the stationary allocation gap, which implies an agricultural-employment odds ratio of $\kappa^{2\gamma}$ in Panel A and $(1-\tau)^{-\gamma}$ in Panel B. Blue and red show the direct composition and wage-mediated effects, respectively; black shows their sum, and the dashed line marks zero. Positive values imply that the next generation continues the initial reallocation out of agriculture; negative values imply partial reversal.

Discussion Intergenerational persistence slows the contemporaneous response to technical change. Workers respond less to relative wages, and reallocation generates a stronger wage feedback. For a common allocation gap, inherited immobility produces the stronger feedback and therefore the smaller contemporaneous response. Across generations, the direct cohort-composition channel dominates under inherited immobility, whereas the stronger wage-mediated channel under inherited productivity can offset or reverse the initial reallocation.

The common stationary allocation, matched allocation gap, pure IGP regimes, symmetric κ , and homothetic demand isolate these mechanisms and their sources. The quantitative counterfactuals retain the common stationary allocation only as a shared initial condition; thereafter, allocations and parental composition evolve endogenously. They also allow both sources of persistence to operate jointly, permit $\kappa_A \neq \kappa_N$, and restore nonhomothetic demand, thereby determining the historical magnitudes of the analytically identified channels.

4. Application: Early 20th Century U.S. Transformation

Here I present the quantitative application to the historical episode of U.S. structural transformation out of agriculture between 1880 and 1940. First, I calibrate the

production-side parameters of the model to obtain the closest match to the historical evolution of agricultural output share, relative price, and real GDP per capita. Then, I identify the sectoral technologies and the relative payoff shifter that exactly replicate the historical evolution of agricultural employment share, relative price, and real GDP per capita. Finally, I simulate the model under counterfactual regimes of intergenerational persistence to quantify how inherited productivity and inherited immobility influence the historical reallocation of workers out of agriculture.

4.1. Production-Side Parameter Calibration

I externally set $\sigma = 0.3$, consistent with the estimates in [Comin, Lashkari and Mestieri \(2021\)](#). I normalize $\Omega_N = 1$ and $\vartheta_A = 1$, calibrate Ω_A from the initial agricultural output share, and choose $\vartheta_N > \vartheta_A$ by minimum distance. I observe the agricultural output share, $\omega_{A,t}$, from [Dennis and İřcan \(2009\)](#), the relative price $p_t \equiv p_{N,t}/p_{A,t}$ from [Alvarez-Cuadrado and Poschke \(2011\)](#), and real GDP per capita, Y_t/P_t , from [Maddison \(1995\)](#) for each year $t \in \{1880, 1900, 1920, 1940\}$. I scale both the relative price and real GDP per capita to be equal to one in 1880, so that the 1880 agricultural output share determines the relative demand shifters.

By normalizing $\Omega_N = 1$, I calibrate Ω_A to match the 1880 agricultural output share:

$$\Omega_A = \frac{\omega_{A,1880}}{\omega_{N,1880}}$$

Then, I choose ϑ_N by minimum distance to the observed agricultural output shares:

$$\hat{\vartheta}_N = \arg \min_{\vartheta_N > 1} \sum_{t \in \{1900, 1920, 1940\}} (\omega_{A,t} - \hat{\omega}_{A,t}(\vartheta_N; \sigma = 0.3))^2.$$

The model-implied relative output share is

$$\frac{\hat{\omega}_{A,t}}{1 - \hat{\omega}_{A,t}} = \Omega_A \left(\frac{p_{N,t}}{p_{A,t}} \right)^{\sigma-1} \left(\frac{Y_t}{P_t} \right)^{(1-\vartheta_N)(1-\sigma)}.$$

As a robustness specification, I also retain a fully external two-sector mapping of the estimates in [Comin, Lashkari and Mestieri \(2021\)](#). Using their agriculture-as-base normalization and mapping manufacturing to the nonagricultural composite gives $\sigma = 0.33$, $\vartheta_A = 1$, and $\vartheta_N = 1/0.33 \approx 3.03$. In both specifications, I continue to normalize $\Omega_N = 1$ and choose Ω_A to match the 1880 U.S. agricultural output share.

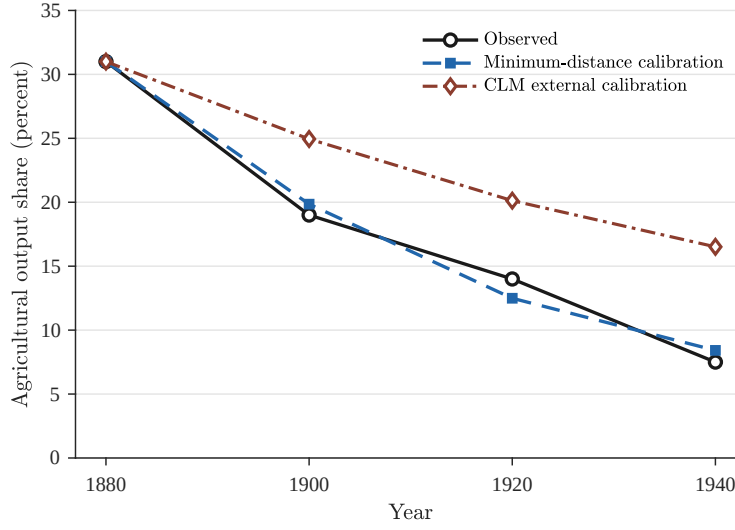


Figure 7: Production-Side Calibration of the Agricultural Output Share

Note: The figure compares the observed agricultural output share with the baseline minimum-distance calibration and the external two-sector mapping of [Comin, Lashkari and Mestieri \(2021\)](#) (CLM). The baseline fixes $\sigma = 0.3$ and $\vartheta_A = 1$ and estimates $\vartheta_N = 4.630$ from the 1900–1940 shares; CLM sets $\sigma = 0.33$, $\vartheta_A = 1$, and $\vartheta_N = 1/0.33 \approx 3.03$, mapping manufacturing to nonagriculture. Both normalize $\Omega_N = 1$ and choose Ω_A to match the 1880 share. Output shares, relative prices, and real GDP per capita are from [Dennis and İřcan \(2009\)](#), [Alvarez-Cuadrado and Poschke \(2011\)](#), and [Maddison \(1995\)](#), respectively.

The minimum-distance estimate of the nonagricultural nonhomotheticity parameter is $\hat{\vartheta}_N = 4.63$. In Figure 7 I plot the observed and model-implied agricultural output shares from the minimum-distance calibration and the external [Comin, Lashkari and Mestieri \(2021\)](#) calibration. By design, both model-implied shares exactly match the observed agricultural output share in 1880. However, for the remaining years, the minimum-distance calibration provides a much better fit than the fully-external calibration. The largest difference for the minimum-distance calibration is 1.5pp in 1920 compared to 9pp in 1940 for the CLM calibration.

Table 5 summarizes the benchmark parameters used in the quantitative analysis and the empirical discipline for each value.

Table 5: Benchmark Model Parameters

Parameter	Value	Empirical discipline
<i>Panel A: Labor-supply parameters</i>		
γ	2.076	Dispersion of log hourly earnings in the 1940 linked-census sample within worker-parent sector cells.
κ_A	1.424	Selection-adjusted hourly-earnings gap between ag-born and nonag-born workers employed in agriculture.
κ_N	1.805	Selection-adjusted hourly-earnings gap between nonag-born and ag-born workers employed in nonagriculture.
τ	0.401	Residual intergenerational sector-choice odds gap after accounting for κ_A and κ_N .
<i>Panel B: Production-side parameters</i>		
Ω_A	0.449	Chosen to match the 1880 agricultural output share.
Ω_N	1	Normalization.
σ	0.300	Set externally, consistent with the estimates in Comin, Lashkari and Mestieri (2021) .
ϑ_A	1	Normalization.
ϑ_N	4.630	Minimum-distance fit to the agricultural output shares in 1900, 1920, and 1940.
<i>Panel C: Initial Conditions</i>		
$\phi_{A,0}$	0.665	Share of ag-born workers in 1880.

Note: Panel A reports the baseline labor-supply parameters: γ and the earnings moments in κ_A and κ_N use 1940 linked-census workers with valid hourly earnings and at least 35 weekly hours, without restricting log earnings; the sector-choice moments in κ_A , κ_N , and τ use the full linked sample. Panel B reports the production-side calibration, and Panel C the initial share of ag-born workers entering in 1880, equal to the 1860 agricultural employment share.

4.2. Inferring Drivers of U.S. Structural Transformation

I exploit the model's structure along with observed agricultural employment shares, relative prices, and real GDP per capita to infer the sectoral technologies and relative payoff shifter that exactly replicate the historical evolution of U.S. structural transformation out of agriculture.

The inversion is period-by-period but recursive because the agricultural employment share in period $t - 1$ determines the fraction of workers in period t who are ag-born. Conditional on this inherited composition and the calibrated parameters, the three observed outcomes in period t identify the three fundamentals $Z_{A,t}$,

$Z_{N,t}$, and $B_{N,t}/B_{A,t}$.

Proposition 3 (Identification by model inversion). *Fix the following model parameters.*

$$\{\gamma, \kappa_A, \kappa_N, \tau, \sigma, \vartheta_A, \vartheta_N, \Omega_A, \Omega_N\}.$$

Assume $\gamma > 1$, $\tau \in [0, 1)$, $\sigma \in (0, 1)$, and strictly positive productivity and demand weights. Normalize $p_{A,t} = 1$ and take the inherited worker composition $\phi_{A,t-1}$ as known. Consider the following interior observed outcomes.

$$\left\{ \phi_{A,t}, p_t \equiv \frac{p_{N,t}}{p_{A,t}}, Q_t \equiv \frac{Y_t}{P_t} \right\} \in (0, 1) \times \mathbb{R}_{++}^2$$

Any observed outcomes in this set that are consistent with the model uniquely identify the period- t fundamentals $\{Z_{A,t}, Z_{N,t}, B_{N,t}/B_{A,t}\} \in \mathbb{R}_{++}^3$.

The result follows from a sequence of closed-form mappings after solving one strictly monotone scalar equation. First, $\phi_{A,t}$ uniquely determines the product of $B_{N,t}/B_{A,t}$ and $w_{N,t}/w_{A,t}$, and hence the sector choices and effective labor supplies of ag-born and nonag-born workers. Second, relative goods demand determines $Z_{N,t}/Z_{A,t}$ from p_t , Q_t , and those effective labor supplies. Third, the level of real GDP pins down the levels of sectoral technologies through the final-good aggregator. Finally, $w_{N,t}/w_{A,t} = p_t Z_{N,t}/Z_{A,t}$ separates the relative payoff shifter from the relative wage. Appendix A.3 provides the complete inversion and proof.

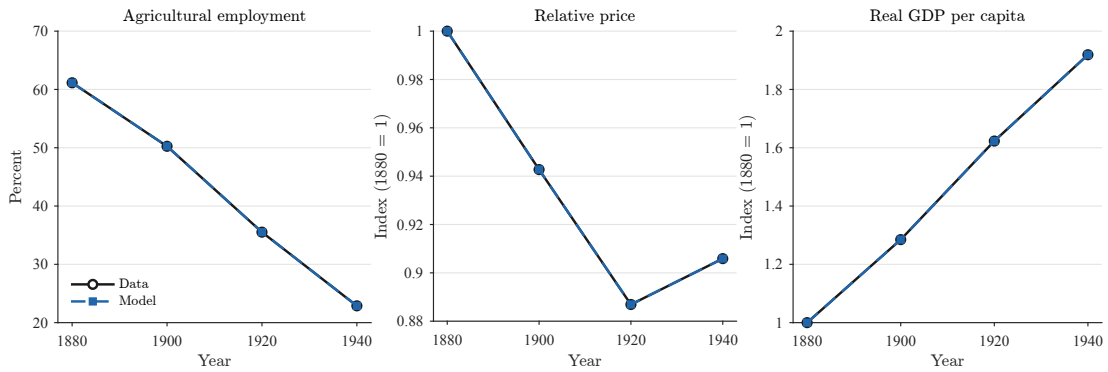


Figure 8: Targeted Moments for Model Inversion

Note: The figure compares the data with the outcomes reproduced by the period-by-period model inversion. Agricultural employment is reported in percent; the relative price, p_N/p_A , and real GDP per capita are indexed to one in 1880. The inversion targets all three series exactly in 1880, 1900, 1920, and 1940, recursively using the previous period's agricultural employment share as the current share of ag-born workers. Employment shares are calculated from the linked censuses; relative prices and real GDP per capita are from [Alvarez-Cuadrado and Poschke \(2011\)](#) and [Maddison \(1995\)](#), respectively.

In Figure 8, I plot the observed agricultural employment share, relative price, and real GDP per capita that are used to identify the sectoral technologies and relative payoff shifter in each year. While the inversion exactly matches the aggregate agricultural employment share, it generally does not match the subgroup employment shares of ag-born and nonag-born workers. I plot observed and model-implied subgroup agricultural employment shares in Figure 9. Most notably, the model implies a much higher agricultural employment share for nonag-born workers in 1880 than is observed in the data.

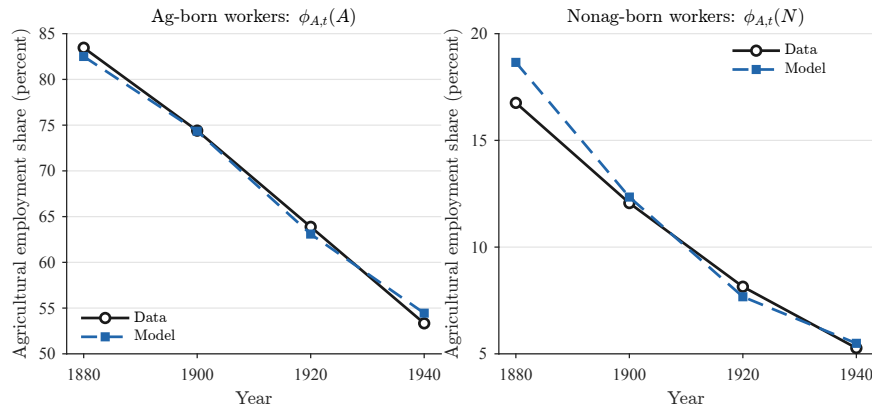


Figure 9: Agricultural Employment Share for Ag-born and Nonag-born Workers

Note: The figure compares observed and model-implied agricultural employment shares for ag-born and nonag-born workers in the linked censuses. These subgroup shares are not targeted: the inversion matches only the aggregate employment share, relative price, and real GDP per capita in each year, using the time-invariant baseline labor-supply parameters estimated from 1940.

This discrepancy occurs mainly because the inherited productivity and immobility are calibrated to match the intergenerational persistence in 1940, which is lower than the persistence in 1880. Within the current model, matching the subgroup paths would require time-varying intergenerational-persistence wedges or an additional mechanism. Without income microdata prior to 1940, estimating such time variation would require maintained assumptions about the evolution of intergenerational persistence that are not supported by the data.

The inferred sectoral technologies and relative payoff shifter, each indexed to their 1880 values, are plotted in Figure 10. I infer dramatic growth in the nonagricultural and, to a lesser extent, the agricultural technology between 1880 and 1940. In particular, the nonagricultural technology is 11 times larger in 1940 than in 1880, while the agricultural technology is about 6.6 times larger. The relative payoff shifter also contributes to the reallocation of workers out of agriculture, but its evolution is much more modest: about 30 percent larger in 1940 than in 1880.

This is consistent with [Alvarez-Cuadrado and Poschke \(2011\)](#) finding that labor pull forces were the dominant driver of U.S. structural transformation out of agriculture in this period.¹⁶

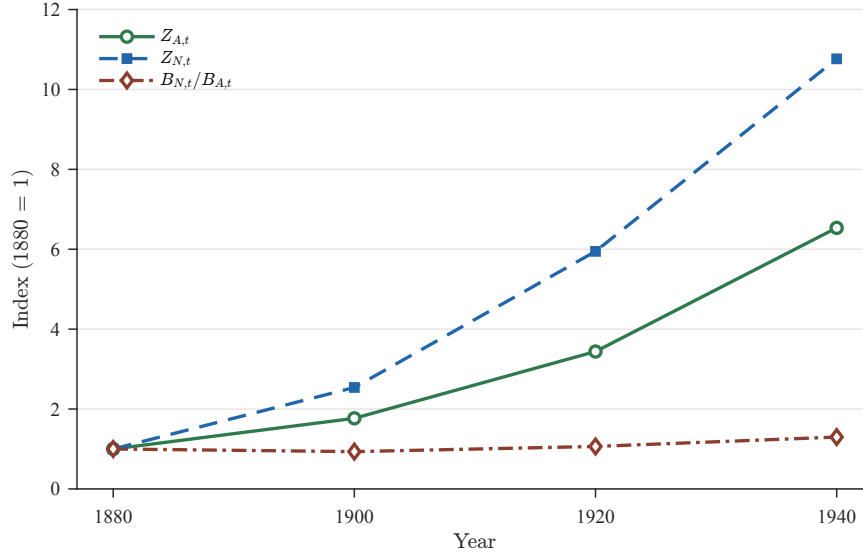


Figure 10: Inferred Drivers of U.S. Structural Transformation, 1880-1940

Note: The figure plots the agricultural and nonagricultural technologies, $Z_{A,t}$ and $Z_{N,t}$, and the relative payoff shifter, $B_{N,t}/B_{A,t}$, inferred from the targeted moments in Figure 8. Each series is indexed to one in 1880; subsequent values are recovered recursively under the baseline parameters in Table 5.

4.3. U.S. Structural Transformation in Counterfactual IGP Regimes

In order to quantify the role of inherited productivity and inherited immobility in shaping U.S. structural transformation, I simulate the model under counterfactual regimes of intergenerational persistence. I consider three counterfactual regimes. First, I remove inherited productivity by setting $\kappa_A = \kappa_N = 1$ while retaining the inherited immobility of ag-born workers by keeping $\tau = 0.401$. Second, I remove inherited immobility by setting $\tau = 0$ while retaining inherited productivity at $\kappa_A = 1.424$ and $\kappa_N = 1.805$. Third, I remove both sources of intergenerational persistence by setting $\kappa_A = \kappa_N = 1$ and $\tau = 0$.

For each counterfactual regime, I re-infer the sectoral technologies and relative payoff shifter that exactly replicate the observed agricultural employment share, relative price, and real GDP per capita in 1880. Then, I simulate the model forward under the counterfactual regime of intergenerational persistence using the inferred

¹⁶ [Alvarez-Cuadrado and Poschke \(2011\)](#) present estimates of relative sectoral productivity growth between 1880 and 1940 equal to 74.4 percent. The implied relative sectoral productivity growth from the inferred sectoral technologies is 64.7 percent.

growth in sectoral technologies and relative payoff shifter depicted in Figure 10. By construction, the counterfactual simulations exactly match the observed outcomes in 1880, but they generally do not match the observed outcomes thereafter.

Table 6: U.S. Worker Reallocation under Counterfactual IGP Regimes, 1880–1940

IGP regime	Worker reallocation	Difference from benchmark	
	Percentage points	Percentage points	Percent
Benchmark	38.3	–	–
No inherited productivity	43.8	5.5	14.4
No inherited immobility	39.4	1.1	2.9
No intergenerational persistence	43.4	5.1	13.3

Note: Worker reallocation is the 1880–1940 decline in the agricultural employment share; differences are relative to the benchmark, in percentage points and as a percent of benchmark reallocation. Each counterfactual is re-anchored to the 1880 employment share, relative price, and real GDP per capita, then simulated using benchmark growth in the inferred technologies and relative payoff shifter.

Table 6 compares the decline in the agricultural employment share between 1880 and 1940 under each counterfactual regime to the benchmark/observed decline. Removing inherited productivity increases worker reallocation out of agriculture by 5.5 percentage points, or 14.4 percent of benchmark reallocation. Removing inherited immobility increases worker reallocation by 1.1 percentage points, or 2.9 percent of benchmark reallocation. Removing both sources of intergenerational persistence increases worker reallocation by 5.1 percentage points, or 13.3 percent of benchmark reallocation.

I next decompose the effect of each counterfactual IGP regime on worker reallocation into the two channels highlighted by the analytical results: workers' contemporaneous response to changing fundamentals and the intergenerational transmission of past reallocation through parental composition. These two channels interact in equilibrium; hence, their measured contributions depend on the order in which they are changed. As such, I use a Shapley decomposition, which assigns each channel its average marginal contribution across the possible orderings. I implement this decomposition within each historical interval and sum the resulting contributions over 1880–1940.

Table 7: Transmission Channels under Counterfactual IGP Regimes

IGP regime	Contemporaneous response	Intergenerational propagation	Total
	Percentage points		
No inherited productivity	2.77	2.74	5.52
No inherited immobility	3.99	−2.87	1.12
No intergenerational persistence	4.36	0.73	5.10

Note: The interval-level Shapley decomposition averages over the two orders in which fundamentals and parental composition change. The contemporaneous channel varies technologies and the relative payoff shifter holding composition fixed; the intergenerational channel varies composition holding those fundamentals fixed. Entries are cumulative 1880–1940 contributions relative to full IGP and sum to the counterfactual reallocation differences in Table 6.

Table 7 shows that removing inherited productivity increases worker reallocation through both channels, with intergenerational propagation accounting for 2.74 of the total 5.52 percentage point effect. Removing inherited immobility generates a large contemporaneous increase in worker reallocation that is mostly offset by the resulting change in parental composition. When I remove both sources of intergenerational persistence, the contemporaneous response increases worker reallocation by 4.36 percentage points, while intergenerational propagation contributes 0.73 percentage points.

I evaluate the robustness of the counterfactual simulation results to alternative calibrations of the production-side and labor-supply parameters. First, I replace the production-side parameters with the external two-sector mapping of [Comin, Lashkari and Mestieri \(2021\)](#) (CLM). Second, I retain the baseline production-side parameters and use the added-fixed-effect estimates of $(\gamma, \kappa_A, \kappa_N, \tau)$. Third, I retain the baseline production-side parameters and use the topcode-adjusted estimates of $(\gamma, \kappa_A, \kappa_N, \tau)$. In each case, I re-infer the historical fundamentals and re-anchor every counterfactual to the common 1880 targets, then repeat the Shapley decomposition of the contemporaneous and intergenerational channels. The results are reported in Table 8, which shows that they are fairly stable across the alternative calibrations. Removing inherited productivity raises reallocation through both channels, whereas removing inherited immobility produces a large contemporaneous response that is substantially offset by intergenerational propagation.

Table 8: Robustness of Transmission Channels under Counterfactual IGP Regimes

IGP regime	Contemporaneous response	Intergenerational propagation	Total
Percentage points			
<i>Panel A: CLM production-side calibration</i>			
No inherited productivity	2.80	2.90	5.70
No inherited immobility	4.05	−2.95	1.10
No intergenerational persistence	4.42	0.83	5.26
<i>Panel B: Added-FE labor-supply calibration</i>			
No inherited productivity	1.31	1.79	3.10
No inherited immobility	2.58	−1.94	0.65
No intergenerational persistence	2.48	0.26	2.73
<i>Panel C: Topcode-adjusted labor-supply calibration</i>			
No inherited productivity	2.78	2.79	5.57
No inherited immobility	4.00	−2.89	1.11
No intergenerational persistence	4.37	0.78	5.15

Note: Each panel repeats the interval-level Shapley decomposition under an alternative calibration. Panel A replaces the production parameters with the CLM mapping: $\sigma = 0.33$, $\vartheta_A = 1$, and $\vartheta_N = 1/0.33$. Panels B and C retain the baseline production parameters and use, respectively, the added-fixed-effect estimates $(\gamma, \kappa_A, \kappa_N, \tau) = (2.293, 1.205, 1.431, 0.341)$ and topcode-adjusted estimates $(2.063, 1.434, 1.809, 0.401)$. Each specification re-infers the historical fundamentals and re-anchors every counterfactual to the common 1880 targets; the two channels sum to the row total.

5. Conclusion

This paper joins two growing literatures: those that study the macroeconomic consequences of intergenerational persistence and those that examine the implications of a less-than-perfectly-elastic labor supply for structural transformation. I develop a theoretical framework in which the sons of farmers may be disproportionately drawn to agricultural work because their parents' sector choice confers either a productive advantage in farming or a broad disinclination toward non-agricultural work. I show that these two sources of intergenerational persistence have distinct implications for structural transformation, and that they can be separately identified using historical microdata on intergenerational occupation choice and earnings. Taking the model to data, I find that intergenerational persistence meaningfully reduced the United States' departure from agriculture in the first half of the 20th century, and that inherited productivity was the primary source of this

reduction.

One direction for future work is to ask whether intergenerational persistence shapes structural transformation in developing economies today. Workers in these economies face substantial barriers to moving across sectors and locations ([Bryan and Morten, 2019](#); [Donovan and Schoellman, 2023](#)), and agricultural occupations are often strongly correlated across generations ([Emran and Shilpi, 2011](#)). The sources of that persistence are likely to matter. Rural insurance networks, for example, can discourage migration even when urban wages are higher ([Munshi and Rosenzweig, 2016](#)); in other settings, occupational persistence may transmit valuable sector-specific skills. Applying the framework to modern linked household data could distinguish these mechanisms and clarify when policies that weaken occupational ties improve the allocation of labor and when they instead erode productive specialization.

A second direction is to incorporate intergenerational persistence into models of adjustment to shorter-run sectoral shocks. The China trade shock caused large and persistent employment losses in exposed U.S. labor markets ([Autor, Dorn and Hanson, 2013](#)). [Caliendo, Dvorkin and Parro \(2019\)](#) show how sectoral and geographic mobility frictions shape the economy's transition after such a shock. The mechanism in this paper adds another source of dynamics: current reallocation changes the parental-sector composition of the next cohort, whose choices may therefore remain connected to the pre-shock allocation. This connection is especially relevant given evidence that the eventual recovery of exposed places came disproportionately from workers who were below working age when the shock began, rather than from incumbent workers changing sectors ([Autor et al., 2025](#)). Embedding inherited productivity and immobility in a dynamic trade model could therefore reveal whether adjustment by new cohorts attenuates a sectoral shock or carries its effects into the next generation.

Intergenerational mobility has become a central concern for researchers and policymakers seeking to understand the persistence of inequality and the scope for policy to expand economic opportunity ([OECD, 2018](#); [Narayan et al., 2018](#)).¹⁷ This paper emphasizes a different consequence of persistence: its effect on an economy's capacity to reallocate workers over long horizons. The results show that

¹⁷Much of this literature measures mobility using income or educational attainment rather than occupation or sector of work; prominent examples include [Black and Devereux \(2011\)](#), [Chetty et al. \(2014\)](#), [Chetty and Hendren \(2018a\)](#), [Chetty and Hendren \(2018b\)](#), and [Chetty et al. \(2020\)](#). Beyond the studies cited earlier, recent work on the determinants of intergenerational occupational mobility includes [Bell et al. \(2019\)](#) and [Greenberg et al. \(2024\)](#).

this effect can be substantial and depends critically on whether persistence reflects inherited productivity or inherited immobility. As artificial intelligence and other technologies reshape the demand for workers across occupations and sectors ([Acemoglu and Restrepo, 2019](#)) in modern economies, forecasts that treat new workers as independent of the previous generation may misstate both the speed and the persistence of labor-market adjustment. Researchers and policymakers anticipating these changes should therefore consider not only whether incumbent workers can move, but also how the skills and attachments inherited by new workers shape where they enter the economy.

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A. Appendix

A.1. Labor Supply Parameter Estimates

This appendix provides additional details and robustness checks for the labor-supply parameter estimates in Section 2. I first establish the conditional variance result used to identify γ and decompose log-odds persistence between the two wedges. I then assess alternative treatments of topcoded earnings and conclude by considering annual and weekly earnings.

A.1.1. Conditional Log-Income Variance

Fix a parental sector k , and write indirect utility in sector j as $V_{i,j,t}(k) = a_{j,t}(k)\epsilon_{i,j,t}$, where $a_{j,t}(k)$ collects the wage, payoff shifter, and applicable inherited wedges. Because $\epsilon_{i,j,t}^{-\gamma}$ is standard exponential,

$$T_{i,j,t}(k) \equiv V_{i,j,t}(k)^{-\gamma} \sim \text{Exp}(a_{j,t}(k)^\gamma)$$

independently across sectors. Maximizing $V_{i,j,t}(k)$ is therefore equivalent to winning an exponential race by minimizing $T_{i,j,t}(k)$. For independent exponentials, the minimum $T_{i,t}^{\min}(k)$ is independent of the identity of the minimizing sector and has rate $\sum_j a_{j,t}(k)^\gamma = \Lambda_t(k)^\gamma$. Thus $V_{i,t}^{\max}(k) = [T_{i,t}^{\min}(k)]^{-1/\gamma}$ is independent of sector choice, and, because log income conditional on (j, k) differs from $\log V_{i,t}^{\max}(k)$ only by a constant,

$$\text{Var}[\log y_{i,j,t}(k) \mid s_{i,t} = j, s_{i,t-1} = k] = \frac{\pi^2}{6\gamma^2}.$$

A.1.2. Decomposing Log-Odds Persistence

Equation (ID-4) decomposes the log intergenerational odds ratio additively as

$$\log \mathcal{O} = \underbrace{\gamma (\log \kappa_A + \log \kappa_N)}_{\text{inherited productivity}} + \underbrace{-\gamma \log(1 - \tau)}_{\text{inherited immobility}}.$$

I define each wedge's contribution to persistence as its component divided by $\log \mathcal{O}$. Using $(\hat{\gamma}, \hat{\kappa}_A, \hat{\kappa}_N, \hat{\tau}) = (2.076, 1.424, 1.805, 0.401)$ and $\hat{\mathcal{O}} = 20.541$ gives

$$\hat{s}_\kappa = \frac{\hat{\gamma}(\log \hat{\kappa}_A + \log \hat{\kappa}_N)}{\log \hat{\mathcal{O}}} = 0.648, \quad \hat{s}_\tau = \frac{-\hat{\gamma} \log(1 - \hat{\tau})}{\log \hat{\mathcal{O}}} = 0.352.$$

Thus inherited productivity accounts for 64.8 percent of observed log-odds persistence and inherited immobility accounts for the remaining 35.2 percent.

A.1.3. Adjustment for Topcoded Earnings

The main text retains the variance-based estimator but checks its sensitivity to replacing each topcoded annual income with a multiple of the \$5,000 censoring threshold. In the final variance-estimation sample, 2,164 of 2,077,253 workers (0.104 percent) are topcoded, and their capped observations account for 0.501 percent of total wage and salary income. I calibrate the multiplier to 2.926 so that these workers instead account for 1.452 percent of total income, the 1940 income share of the top 0.1 percent reported by [Atkinson, Piketty and Saez \(2011\)](#). I then reconstruct hourly earnings and reestimate γ from the residual variance on the fixed main-text sample.

Table 9 shows that the estimate changes little over a broad range of multipliers. At the calibrated value, $\hat{\gamma}$ declines from 2.076 to 2.063, while the quantile RMSE rises only from 0.337 to 0.340.

Table 9: Sensitivity to the Topcode Adjustment

Topcode multiplier	Topcoded income share (%)	$\hat{\gamma}$	Observations	Quantile RMSE
1.00	0.501	2.076	2,077,253	0.337
1.25	0.625	2.074	2,077,253	0.338
1.50	0.749	2.072	2,077,253	0.338
1.75	0.873	2.070	2,077,253	0.339
2.00	0.997	2.068	2,077,253	0.339
2.93	1.452	2.063	2,077,253	0.340

Note: Each row replaces topcoded annual income with the stated multiple of \$5,000 before constructing hourly earnings and reestimating γ . The 2.93 multiplier matches the 1.452 percent target share; quantile RMSE is measured over the 1st–99th residual percentiles.

A.1.4. Censored Maximum-Likelihood Estimation of γ

The main specification infers γ from the residual variance of log hourly earnings. As a robustness check, I instead impose the full model-implied log-Fréchet distribution and account explicitly for the 1940 Census topcode. Let x_i denote log hourly earnings and let μ_{jk} be an unrestricted location parameter for workers employed in sector j whose fathers worked in sector k . For an uncensored observation, the

likelihood contribution is

$$\log f(x_i | \mu_{jk}, \gamma) = \log \gamma - \gamma(x_i - \mu_{jk}) - \exp[-\gamma(x_i - \mu_{jk})].$$

For a topcoded worker, annual earnings are known only to exceed \$5,000. Because workers report different weeks and hours, the corresponding log-hourly censoring threshold is observation-specific:

$$c_i = \log\left(\frac{5,000}{\text{weeks}_i \times \text{hours}_i}\right).$$

The censored contribution is therefore

$$\log[1 - F(c_i | \mu_{jk}, \gamma)] = \log\{1 - \exp[-\exp(-\gamma(c_i - \mu_{jk}))]\}.$$

I jointly estimate the four μ_{jk} parameters and the common shape parameter, imposing $\gamma > 1$ through the transformation $\gamma = 1 + \exp(\tilde{\gamma})$. Table 10 compares the result with the baseline variance estimator. For numerical stability, this estimator alone restricts log hourly earnings to $[-4, 2]$; all variance-based estimates use the unrestricted sample described in the main text. The MLE produces a substantially lower estimate, 1.406, and a higher quantile RMSE, 0.554 rather than 0.337. Although its likelihood correctly treats the right-censored observations, it imposes the log-Fréchet distribution on the entire sample and is therefore sensitive to departures from that distribution, including in the much larger uncensored lower tail. Given its poorer distributional fit, I retain it only as an appendix robustness check.

Table 10: Variance and Censored-Likelihood Estimates of γ

	Baseline variance	Censored MLE
$\hat{\gamma}$	2.076	1.406
Observations	2,077,253	2,065,302
Quantile RMSE (1st–99th percentiles)	0.337	0.554

Note: The MLE jointly estimates four sector-by-parental-sector locations and a common γ , treating topcoded observations as right-censored at their individual hourly thresholds and restricting log hourly earnings to $[-4, 2]$. The baseline uses the unrestricted 2,077,253-observation sample; the MLE uses 2,065,302 observations. Quantile RMSE evaluates both estimates against the unrestricted sample’s 1st–99th residual percentiles.

Table 11 carries both treatments of topcoding through the wedge calculations. The calibrated adjustment barely changes the wedges, whereas the lower MLE estimate implies larger inherited productivity wedges.

Table 11: Inferred Wedges under Alternative Estimates of γ

	Baseline variance	Topcode adjusted	Censored MLE
$\hat{\gamma}$	2.076	2.063	1.406
$\hat{\kappa}_A$	1.424	1.434	2.421
$\hat{\kappa}_N$	1.805	1.809	2.123
$\hat{\tau}$	0.401	0.401	0.401

Note: Columns vary $\hat{\gamma}$ while holding the baseline income and sector-choice moments fixed, isolating the effect of the estimator. The topcode-adjusted estimate uses the calibrated 2.926 multiplier; the censored MLE uses the likelihood above.

A.1.5. Alternative Earnings Measures

Hourly earnings are the closest empirical counterpart to the model's wage per efficiency unit, but they inherit measurement error from both weeks and hours. Annual and weekly earnings provide useful checks on whether this additional measurement affects the estimated dispersion of earnings. For comparability, Table 12 estimates all three concepts on the common preferred sample used in the main text.

Table 12: Variance-Based Estimates of the Fréchet Shape Parameter by Income Measure

	Annual		Weekly		Hourly	
	Baseline	Added FE	Baseline	Added FE	Baseline	Added FE
$\hat{\gamma}$	2.187	2.422	2.214	2.448	2.076	2.293
Residual SD of log wages	0.586	0.530	0.579	0.524	0.618	0.559
Observations	2,077,253	2,077,253	2,077,253	2,077,253	2,077,253	2,077,253
Topcoded observations	2,164	2,164	2,164	2,164	2,164	2,164
Topcoded share (%)	0.104	0.104	0.104	0.104	0.104	0.104
Earnings regression R^2	0.307	0.435	0.306	0.433	0.341	0.460

Note: Estimates use $\hat{\gamma} = \pi/(\sqrt{6}\hat{\sigma})$ after removing worker-sector-by-parental-sector effects; added-FE columns also control for adult commuting zone, childhood schooling, and age. Weekly earnings divide annual wage and salary income by weeks worked, and hourly earnings additionally divide by usual weekly hours. All columns use the common unrestricted 2,077,253-observation sample defined in Table 3.

The estimates are similar across earnings concepts: the baseline estimates range from 2.076 for hourly earnings to 2.214 for weekly earnings, while the added-fixed-effect estimates range from 2.293 to 2.448. Figure 11 shows the corresponding distributional fits.

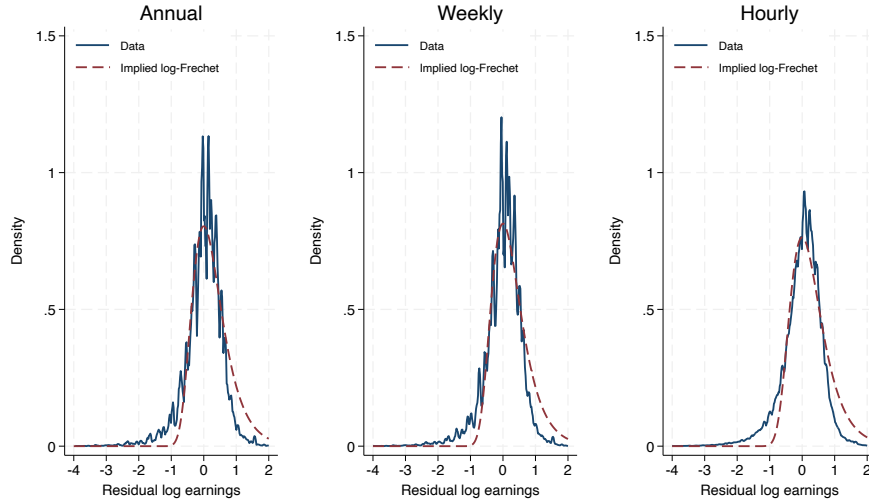


Figure 11: Residual Earnings Distributions and Implied Log-Fréchet Densities by Income Measure

Note: The solid lines plot kernel densities of residual log earnings after removing worker-sector-by-parental-sector cell means. Dashed lines plot the log-Fréchet densities implied by the baseline estimates in Table 12. Panels use the common unrestricted estimation sample and display residuals between -4 and 2 .

Finally, Table 13 carries each variance-based estimate of γ through the wedge calculations. The estimated inherited productivity and immobility wedges remain economically important across earnings concepts and control sets, although their magnitudes vary with the dispersion estimate and adjusted income gaps.

Table 13: Inherited Productivity and Immobility Wedges by Income Measure

	Annual earnings		Weekly earnings		Hourly earnings	
	Baseline	Added FE	Baseline	Added FE	Baseline	Added FE
<i>Panel A: Inferred wedges</i>						
Inherited agricultural productivity, $\hat{\kappa}_A$	1.669	1.463	1.559	1.358	1.424	1.205
Inherited nonagricultural productivity, $\hat{\kappa}_N$	1.814	1.467	1.756	1.419	1.805	1.431
Inherited mobility wedge, $\hat{\tau}$	0.240	0.138	0.301	0.218	0.401	0.341
<i>Panel B: Mean earnings</i>						
Agriculture, ag-born: $\hat{y}_A(A)$	258.44	275.97	7.23	7.73	0.16	0.17
Agriculture, nonag-born: $\hat{y}_A(N)$	446.05	375.25	13.19	11.23	0.34	0.29
Nonagriculture, ag-born: $\hat{y}_N(A)$	679.72	744.97	17.26	18.95	0.46	0.51
Nonagriculture, nonag-born: $\hat{y}_N(N)$	892.13	873.54	22.02	21.54	0.60	0.58
<i>Panel C: Intergenerational persistence</i>						
Sector-choice odds ratio	20.541	9.095	20.541	9.095	20.541	9.095

Note: Each column uses the corresponding estimate of γ in Table 12. Added-FE columns adjust income and sector-choice moments for adult commuting zone, childhood schooling, and age. Earnings moments use the relevant income sample; all sector-choice moments entering κ_j , τ , and Panel C use the full linked sample, as in Table 4.

A.2. Characterizing the Influence of Structural Transformation

A.2.1. Contemporaneous Transmission of Agricultural Technology Shock

The main text reduces the contemporaneous equilibrium to a relative headcount labor-supply schedule,

$$d \log \frac{\phi_{N,t}}{\phi_{A,t}} = \mathcal{E} d \log \frac{w_{N,t}}{w_{A,t}},$$

and an inverse relative labor-demand schedule expressed in terms of headcount allocations,

$$d \log \frac{w_{N,t}}{w_{A,t}} = \mathcal{D}^{Z_A} d \log Z_{A,t} + \mathcal{H} d \log \frac{\phi_{N,t}}{\phi_{A,t}}.$$

The three objects in these schedules have economically distinct ceteris-paribus interpretations. The technology elasticity $\mathcal{D}^{Z_A}$ shifts inverse relative labor demand while holding sectoral effective labor supplies fixed:

$$\mathcal{D}^{Z_A} \equiv \left. \frac{\partial \log(w_{N,t}/w_{A,t})}{\partial \log Z_{A,t}} \right|_{d \log H_{A,t} = d \log H_{N,t} = 0}.$$

The worker-allocation elasticity $\mathcal{E}$ holds cohort composition and the relative payoff shifter fixed. The feedback elasticity $\mathcal{H}$ holds technologies fixed and traces how effective labor, production, and relative goods prices respond along the same within-cohort worker-allocation margin underlying $\mathcal{E}$. Solving the two schedules

and using $d\phi_{A,t} = -\phi_{A,t}(1 - \phi_{A,t})d\log(\phi_{N,t}/\phi_{A,t})$ gives

$$\frac{d\phi_{A,t}}{d\log Z_{A,t}} = -\phi_{A,t}(1 - \phi_{A,t})\frac{\mathcal{E}}{1 - \mathcal{E}\mathcal{H}}\mathcal{D}^{Z_A}.$$

Below I derive each schedule from the full equilibrium system. I begin with the benchmark version of the economy without intergenerational persistence.

Supporting Derivations without IGP The agricultural technology innovation provides a direct shock to labor demand:

$$d\log \frac{w_{N,t}}{w_{A,t}} = d\log Z_{N,t} - d\log Z_{A,t} + d\log p_t$$

It also provides an indirect shock to labor demand through the relative price:

$$d\log p_t = -\frac{1}{\sigma} d\log \frac{Q_{N,t}}{Q_{A,t}} + \frac{(\vartheta_N - \vartheta_A)(1 - \sigma)}{\sigma} d\log Q_t$$

given both the relative sectoral and aggregate output respond to the agricultural technology shock:

$$d\log Q_{A,t} = d\log Z_{A,t} + d\log H_{A,t}$$

$$d\log Q_{N,t} = d\log Z_{N,t} + d\log H_{N,t}$$

$$d\log Q_t = \frac{\omega_{A,t}}{\omega_{A,t}\vartheta_A + \omega_{N,t}\vartheta_N} d\log Q_{A,t} + \frac{\omega_{N,t}}{\omega_{A,t}\vartheta_A + \omega_{N,t}\vartheta_N} d\log Q_{N,t}$$

Hence, holding the effective labor supply fixed, the contemporaneous response of the relative wage to the agricultural technology shock is given by:

$$\mathcal{D}^{Z_A} \equiv \frac{d\log(w_{N,t}/w_{A,t})}{d\log Z_{A,t}} \Big|_{dH_{j,t}=0} = \frac{1 - \sigma}{\sigma} \cdot \frac{\vartheta_N}{\omega_{A,t}\vartheta_A + \omega_{N,t}\vartheta_N}$$

The relative worker allocation satisfies:

$$d\log \frac{\phi_{N,t}}{\phi_{A,t}} = \gamma \cdot \left[d\log \frac{w_{N,t}}{w_{A,t}} + d\log \frac{B_{N,t}}{B_{A,t}} \right]$$

and the effective labor supply satisfies:

$$d \log H_{A,t} = \frac{\gamma - 1}{\gamma} \cdot d \log \phi_{A,t} \quad \text{and} \quad d \log H_{N,t} = -\frac{\phi_{A,t}}{1 - \phi_{A,t}} \cdot d \log H_{A,t}$$

Substituting these into the relative output equation yields:

$$d \log \frac{Q_{N,t}}{Q_{A,t}} = -\frac{1}{1 - \phi_{A,t}} \cdot \frac{\gamma - 1}{\gamma} \cdot d \log \phi_{A,t}$$

while substituting into the aggregate output equation yields:

$$d \log Q_t = \left[\frac{\omega_{A,t} - \phi_{A,t}}{(1 - \phi_{A,t})(\omega_{A,t}\vartheta_A + \omega_{N,t}\vartheta_N)} \right] \cdot \frac{\gamma - 1}{\gamma} \cdot d \log \phi_{A,t}$$

Hence, the relative wage responds to worker reallocation according to:

$$d \log \frac{w_{N,t}}{w_{A,t}} = -\frac{\gamma - 1}{\gamma} \cdot \left[\frac{1}{\sigma} + \frac{(\vartheta_N - \vartheta_A)(1 - \sigma)}{\sigma} \cdot \frac{\omega_{A,t} - \phi_{A,t}}{\omega_{A,t}\vartheta_A + \omega_{N,t}\vartheta_N} \right] \cdot d \log \frac{\phi_{N,t}}{\phi_{A,t}}$$

Define:

$$\mathcal{E} \equiv \frac{\partial \log(\phi_{N,t}/\phi_{A,t})}{\partial \log(w_{N,t}/w_{A,t})} \Big|_{d \log(B_{N,t}/B_{A,t})=0} = \gamma.$$

Similarly,

$$\begin{aligned} \mathcal{H} &\equiv \frac{d \log(w_{N,t}/w_{A,t})}{d \log(\phi_{N,t}/\phi_{A,t})} \Big|_{d \log Z_{A,t}=d \log Z_{N,t}=0} \\ &= -\frac{\gamma - 1}{\gamma} \cdot \left[\frac{1}{\sigma} + \frac{(\vartheta_N - \vartheta_A)(1 - \sigma)}{\sigma} \cdot \frac{\omega_{A,t} - \phi_{A,t}}{\omega_{A,t}\vartheta_A + \omega_{N,t}\vartheta_N} \right] \end{aligned}$$

Then, the relative worker allocation response to a labor demand shock originating from the agricultural technology shock is given by:

$$\begin{aligned} d \log \frac{\phi_{N,t}}{\phi_{A,t}} &= \mathcal{E} \cdot d \log \frac{w_{N,t}}{w_{A,t}} \\ &= \mathcal{E} \cdot \left[\mathcal{D}^{Z_A} d \log Z_{A,t} + \mathcal{H} \cdot d \log \frac{\phi_{N,t}}{\phi_{A,t}} \right] \\ &= \frac{\mathcal{E}}{1 - \mathcal{E} \cdot \mathcal{H}} \cdot \mathcal{D}^{Z_A} \cdot d \log Z_{A,t} \end{aligned}$$

Hence, as noted above, I arrive at:

$$\frac{d\phi_{A,t}}{d \log Z_{A,t}} = -\phi_{A,t}(1 - \phi_{A,t}) \cdot \frac{\mathcal{E}}{1 - \mathcal{E} \cdot \mathcal{H}} \cdot \mathcal{D}^{Z_A}$$

This expression will hold even in the presence of intergenerational persistence, but the underlying closed-form solutions for $\mathcal{E}$ and $\mathcal{H}$ will change.

Worker-allocation elasticity with IGP Let $\mu_{k,t} \equiv \phi_{k,t-1}$ denote the share of workers born to sector k and define

$$S_{0,t} \equiv \sum_{k \in \{A,N\}} \mu_{k,t} \phi_{A,t}(k) \phi_{N,t}(k).$$

Equation (1) implies

$$\frac{\partial \phi_{A,t}(k)}{\partial \log(w_{N,t}/w_{A,t})} = -\gamma \phi_{A,t}(k) \phi_{N,t}(k).$$

Holding cohort composition fixed and aggregating across parental sectors therefore gives

$$\frac{\partial \phi_{A,t}}{\partial \log(w_{N,t}/w_{A,t})} = -\gamma S_{0,t}.$$

Because $\phi_{N,t} = 1 - \phi_{A,t}$,

$$\mathcal{E} = \gamma \frac{S_{0,t}}{\phi_{A,t} \phi_{N,t}}.$$

Under stationarity, $\phi_{A,t-1} = \phi_{A,t}$, equation (3) implies

$$\frac{S_{0,t}}{\phi_{A,t} \phi_{N,t}} = 1 - [\phi_{A,t}(A) - \phi_{A,t}(N)]^2.$$

Thus the worker-allocation elasticity in an economy with IGP is

$$\mathcal{E} = \gamma \{1 - [\phi_{A,t}(A) - \phi_{A,t}(N)]^2\}.$$

General-equilibrium feedback with IGP To derive $\mathcal{H}$ without imposing homothetic demand, first define the normalized sector-specific effective-labor responses

$$\begin{aligned}\Psi_{A,t} &\equiv \frac{\phi_{A,t}\phi_{N,t}}{S_{0,t}} \sum_{k \in \{A,N\}} \mu_{k,t}\phi_{A,t}(k)\phi_{N,t}(k) \frac{h_{A,t}(k)}{H_{A,t}}, \\ \Psi_{N,t} &\equiv \frac{\phi_{A,t}\phi_{N,t}}{S_{0,t}} \sum_{k \in \{A,N\}} \mu_{k,t}\phi_{A,t}(k)\phi_{N,t}(k) \frac{h_{N,t}(k)}{H_{N,t}}.\end{aligned}$$

Differentiating equation (2) gives

$$d \log h_{j,t}(k) = -\frac{1}{\gamma} d \log \phi_{j,t}(k).$$

Combining this selection response with the subcohort allocation response gives

$$\begin{aligned}\frac{\partial \log H_{A,t}}{\partial \log(w_{N,t}/w_{A,t})} &= -(\gamma - 1) \sum_k \mu_{k,t}\phi_{A,t}(k)\phi_{N,t}(k) \frac{h_{A,t}(k)}{H_{A,t}}, \\ \frac{\partial \log H_{N,t}}{\partial \log(w_{N,t}/w_{A,t})} &= (\gamma - 1) \sum_k \mu_{k,t}\phi_{A,t}(k)\phi_{N,t}(k) \frac{h_{N,t}(k)}{H_{N,t}}.\end{aligned}$$

Dividing by the worker-allocation elasticity derived above yields

$$\begin{aligned}\frac{d \log H_{A,t}}{d \log(\phi_{N,t}/\phi_{A,t})} &= -\frac{\gamma - 1}{\gamma} \Psi_{A,t}, \\ \frac{d \log H_{N,t}}{d \log(\phi_{N,t}/\phi_{A,t})} &= \frac{\gamma - 1}{\gamma} \Psi_{N,t}.\end{aligned}$$

Holding technologies fixed, sectoral production implies $d \log Q_{j,t} = d \log H_{j,t}$. Hence

$$\frac{d \log(Q_{N,t}/Q_{A,t})}{d \log(\phi_{N,t}/\phi_{A,t})} = \frac{\gamma - 1}{\gamma} (\Psi_{N,t} + \Psi_{A,t}),$$

and the final-good aggregator implies

$$\frac{d \log Q_t}{d \log(\phi_{N,t}/\phi_{A,t})} = \frac{\gamma - 1}{\gamma} \frac{\omega_{N,t}\Psi_{N,t} - \omega_{A,t}\Psi_{A,t}}{\omega_{A,t}\vartheta_A + \omega_{N,t}\vartheta_N}.$$

Substitution into the relative-wage equation gives the full nonhomothetic feedback elasticity:

$$\mathcal{H} = \frac{\gamma - 1}{\gamma} \left\{ -\frac{\Psi_{A,t} + \Psi_{N,t}}{\sigma} + \frac{(\vartheta_N - \vartheta_A)(1 - \sigma)}{\sigma} \frac{\omega_{N,t}\Psi_{N,t} - \omega_{A,t}\Psi_{A,t}}{\omega_{A,t}\vartheta_A + \omega_{N,t}\vartheta_N} \right\}.$$

The first term is the relative-price feedback; the second is the aggregate-income feedback generated by nonhomothetic demand. In the homothetic case, $\vartheta_A = \vartheta_N = 1$, the second term vanishes and

$$\mathcal{H} = -\frac{1}{\sigma} \frac{\gamma - 1}{\gamma} (\Psi_{A,t} + \Psi_{N,t}).$$

In the benchmark economy, $\Psi_{A,t} = \phi_{N,t}$ and $\Psi_{N,t} = \phi_{A,t}$. Thus the nonhomothetic expression nests the earlier benchmark result, while its homothetic specialization reduces to $\mathcal{H}^{\text{bench}} = -(\gamma - 1)/(\gamma\sigma)$.

Proof of Lemma 1 Recall, Lemma 1 states:

The elasticity of worker allocation with respect to the relative wage is strictly smaller in the presence of intergenerational persistence regardless of whether that persistence arises from inherited productivity or inherited immobility.

$$0 < \mathcal{E}^{\text{IGP-}\kappa} = \mathcal{E}^{\text{IGP-}\tau} < \mathcal{E}^{\text{bench}} = \gamma$$

Proof. The stationary expression for $\mathcal{E}$ shows that it depends on IGP only through the allocation gap $\Delta_t \equiv \phi_{A,t}(A) - \phi_{A,t}(N)$. In the benchmark economy, $\Delta_t = 0$, so $\mathcal{E}^{\text{bench}} = \gamma$. Each IGP economy has $0 < \Delta_t < 1$, implying $0 < \gamma(1 - \Delta_t^2) < \gamma$. Finally, the two IGP economies are constructed to have the same allocation gap and therefore have the same elasticity. $\square$

Proof of Lemma 2 Recall, Lemma 2 states:

The magnitude of the general-equilibrium feedback of worker reallocation into the relative wage is strictly larger in the presence of intergenerational persistence, and it is largest when that persistence arises from inherited immobility:

$$\mathcal{H}^{\text{IGP-}\tau} < \mathcal{H}^{\text{IGP-}\kappa} \leq \mathcal{H}^{\text{bench}} = -\frac{1}{\sigma} \frac{\gamma - 1}{\gamma},$$

where the weak inequality is an equality only when $\phi_{A,t-1} = 0.5$.

Proof. Under homothetic demand, the expression for $\mathcal{H}$ above shows that its ordering is the reverse of the ordering of $\Psi_{A,t} + \Psi_{N,t}$. The benchmark values $\Psi_{A,t} = \phi_{N,t}$

and $\Psi_{N,t} = \phi_{A,t}$ imply $\Psi_{A,t} + \Psi_{N,t} = 1$ and hence

$$\mathcal{H}^{\text{bench}} = -\frac{1}{\sigma} \frac{\gamma - 1}{\gamma}.$$

For the two IGP economies, equation (1) gives

$$\frac{\phi_{A,t}(A)/\phi_{N,t}(A)}{\phi_{A,t}(N)/\phi_{N,t}(N)} = \begin{cases} \kappa^{2\gamma} & \text{in the IGP-}\kappa \text{ economy} \\ (1 - \tau)^{-\gamma} & \text{in the IGP-}\tau \text{ economy.} \end{cases}$$

The common allocation gap and common stationary aggregate allocation therefore pin down the wedges being compared, but the ordering does not follow from the odds ratio alone. To make the remaining step explicit, let

$$m_t \equiv \phi_{A,t} = \phi_{A,t-1}, \quad \Delta_t \equiv \phi_{A,t}(A) - \phi_{A,t}(N), \quad q \equiv \frac{\gamma - 1}{\gamma}.$$

Stationarity implies that the common subcohort allocations in the two IGP economies are

$$a_t \equiv \phi_{A,t}(A) = m_t + (1 - m_t)\Delta_t, \quad b_t \equiv \phi_{A,t}(N) = m_t(1 - \Delta_t).$$

Write $\bar{a}_t \equiv 1 - a_t$ and $\bar{b}_t \equiv 1 - b_t$. The odds-ratio restriction for the IGP- κ economy becomes

$$\kappa^{2\gamma} = \frac{a_t \bar{b}_t}{\bar{a}_t b_t}.$$

Substituting equation (2) into the definitions of $\Psi_{A,t}$ and $\Psi_{N,t}$ gives

$$\begin{aligned} \Psi_{A,t}(\kappa) &= \frac{1}{1 - \Delta_t^2} \frac{m_t \kappa a_t^q \bar{a}_t + (1 - m_t) b_t^q \bar{b}_t}{m_t \kappa a_t^q + (1 - m_t) b_t^q}, \\ \Psi_{N,t}(\kappa) &= \frac{1}{1 - \Delta_t^2} \frac{m_t a_t \bar{a}_t^q + (1 - m_t) \kappa b_t \bar{b}_t^q}{m_t \bar{a}_t^q + (1 - m_t) \kappa \bar{b}_t^q}. \end{aligned}$$

The IGP- τ expressions are the same formulas evaluated at $\kappa = 1$, because inherited immobility affects the common allocations a_t and b_t but does not directly augment effective labor.

Since $a_t > b_t$, we have $\bar{a}_t < \bar{b}_t$. The fraction defining $\Psi_{A,t}(\kappa)$ is a weighted average of $\bar{a}_t$ and $\bar{b}_t$; raising κ places more weight on the smaller value $\bar{a}_t$. Similarly, the fraction defining $\Psi_{N,t}(\kappa)$ is a weighted average of a_t and b_t ; raising κ places more

weight on the smaller value b_t . Because $\kappa > 1$, both comparisons are strict:

$$\Psi_{A,t}^{\text{IGP-}\tau} > \Psi_{A,t}^{\text{IGP-}\kappa}, \quad \Psi_{N,t}^{\text{IGP-}\tau} > \Psi_{N,t}^{\text{IGP-}\kappa}.$$

For the comparison with the benchmark, substitute the odds restriction into the preceding expressions for $\Psi_{A,t}(\kappa)$ and $\Psi_{N,t}(\kappa)$ and put the two sectoral terms over a common denominator. After cancellation, the sign of $\Psi_{A,t}^{\text{IGP-}\kappa} + \Psi_{N,t}^{\text{IGP-}\kappa} - 1$ is the sign of $(2m_t - 1)^2$. Consequently,

$$(\Psi_{A,t} + \Psi_{N,t})^{\text{IGP-}\tau} > (\Psi_{A,t} + \Psi_{N,t})^{\text{IGP-}\kappa} \geq 1.$$

The second inequality is strict unless $m_t = 0.5$; at that allocation, the two symmetric inherited-productivity terms offset exactly. Multiplication by the negative factor $-(\gamma - 1)/(\gamma\sigma)$ in the homothetic expression for $\mathcal{H}$ reverses the inequalities and gives the stated ordering. $\square$

Proof of Proposition 1 Recall, Proposition 1 states:

Improvements in agricultural technology induce a reallocation of workers out of agriculture, but the magnitude of that reallocation is smaller in the presence of intergenerational persistence:

$$\left(\frac{d\phi_{A,t}}{d \log Z_{A,t}} \right)^{\text{bench}} < \left(\frac{d\phi_{A,t}}{d \log Z_{A,t}} \right)^{\text{IGP-}\kappa} < \left(\frac{d\phi_{A,t}}{d \log Z_{A,t}} \right)^{\text{IGP-}\tau} < 0.$$

Proof. The definitions of $\mathcal{D}^{Z_A}$, $\mathcal{E}$, and $\mathcal{H}$ imply:

$$\frac{d \log(\phi_{N,t}/\phi_{A,t})}{d \log Z_{A,t}} = \mathcal{E} \cdot \left[\mathcal{D}^{Z_A} + \mathcal{H} \cdot \frac{d \log(\phi_{N,t}/\phi_{A,t})}{d \log Z_{A,t}} \right].$$

Solving this expression and using $d\phi_{A,t} = -\phi_{A,t}(1 - \phi_{A,t})d \log(\phi_{N,t}/\phi_{A,t})$ gives:

$$\frac{d\phi_{A,t}}{d \log Z_{A,t}} = -\phi_{A,t}(1 - \phi_{A,t}) \cdot \frac{\mathcal{E}}{1 - \mathcal{E}\mathcal{H}} \cdot \mathcal{D}^{Z_A}.$$

The response is negative because $\mathcal{D}^{Z_A} > 0$, $\mathcal{E} > 0$, and $\mathcal{H} < 0$. The three economies have the same initial aggregate allocation and the same $\mathcal{D}^{Z_A}$, so their responses can

be ordered by $\mathcal{E}/(1 - \mathcal{E}\mathcal{H})$. This term can be written as:

$$\frac{\mathcal{E}}{1 - \mathcal{E}\mathcal{H}} = \frac{1}{1/\mathcal{E} - \mathcal{H}}.$$

Lemma 1 implies that $1/\mathcal{E}$ is smallest in the benchmark economy and equal across the two IGP economies, while Lemma 2 implies that $-\mathcal{H}$ is smallest in the benchmark economy and largest in the IGP- τ economy. Therefore:

$$\left(\frac{\mathcal{E}}{1 - \mathcal{E}\mathcal{H}}\right)^{\text{bench}} > \left(\frac{\mathcal{E}}{1 - \mathcal{E}\mathcal{H}}\right)^{\text{IGP-}\kappa} > \left(\frac{\mathcal{E}}{1 - \mathcal{E}\mathcal{H}}\right)^{\text{IGP-}\tau} > 0.$$

Multiplying by the common negative term $-\phi_{A,t}(1 - \phi_{A,t})\mathcal{D}^{Z_A}$ reverses these inequalities and proves Proposition 1. $\square$

A.2.2. Dynamic Transmission of Past Reallocation via Parent Composition

With intergenerational persistence, past worker reallocation can affect future worker allocation through the composition of the parent generation. As in the main text, I separate this composition change into two initial shifts: a direct shift in aggregate headcounts and an indirect shift in the relative wage through effective labor supply. The direct shift follows from the aggregate worker allocation in the next period:

$$\phi_{A,t+1} = \phi_{A,t+1}(A)\phi_{A,t} + \phi_{A,t+1}(N)\phi_{N,t}$$

Hence, I have:

$$\mathcal{F}^{\text{direct}} \equiv \left. \frac{\partial \phi_{A,t+1}}{\partial \phi_{A,t}} \right|_{d\phi_{j,t+1}(k)=0} = \phi_{A,t+1}(A) - \phi_{A,t+1}(N).$$

The indirect shift operates through the effective labor supplied to each sector, which in turn affects the relative wage and the next-period subcohort allocations. The indirect effect is given by:

$$\begin{aligned} \mathcal{F}^{H_A} &\equiv \left. \frac{\partial H_{A,t+1}}{\partial \phi_{A,t}} \right|_{d\phi_{j,t+1}(k)=0} = \phi_{A,t+1}(A)h_{A,t+1}(A) - \phi_{A,t+1}(N)h_{A,t+1}(N), \\ \mathcal{F}^{H_N} &\equiv \left. \frac{\partial H_{N,t+1}}{\partial \phi_{A,t}} \right|_{d\phi_{j,t+1}(k)=0} = \phi_{N,t+1}(A)h_{N,t+1}(A) - \phi_{N,t+1}(N)h_{N,t+1}(N). \end{aligned}$$

Thus the composition-induced relative-wage response, holding the period- $t + 1$ subcohort allocations fixed, is

$$\begin{aligned}\mathcal{F}^{indirect} &\equiv \left. \frac{\partial \log(w_{N,t+1}/w_{A,t+1})}{\partial \phi_{A,t}} \right|_{d\phi_{j,t+1}(k)=0} \\ &= \frac{1}{\sigma} \left(\frac{\mathcal{F}^{H_A}}{H_{A,t+1}} - \frac{\mathcal{F}^{H_N}}{H_{N,t+1}} \right) \\ &\quad + \frac{(\vartheta_N - \vartheta_A)(1 - \sigma)}{\sigma} \frac{\omega_{A,t+1} \mathcal{F}^{H_A}/H_{A,t+1} + \omega_{N,t+1} \mathcal{F}^{H_N}/H_{N,t+1}}{\omega_{A,t+1} \vartheta_A + \omega_{N,t+1} \vartheta_N}.\end{aligned}$$

The first term is the relative-price response to the composition-induced change in relative effective labor. The second term is the corresponding aggregate-income response under nonhomothetic demand.

To incorporate the contemporaneous equilibrium response at $t + 1$, let

$$v_{t+1} \equiv \frac{d \log(w_{N,t+1}/w_{A,t+1})}{d \phi_{A,t}}.$$

The dynamic equilibrium can then be reduced to the same two equations used in the main text. Differentiating the aggregate next-period worker allocation gives the headcount response

$$\frac{d \phi_{A,t+1}}{d \phi_{A,t}} = \mathcal{F}^{direct} - \phi_{A,t+1} \phi_{N,t+1} \mathcal{E} v_{t+1},$$

while the relative-wage response satisfies

$$v_{t+1} = \mathcal{F}^{indirect} + \mathcal{H} \mathcal{E} v_{t+1}.$$

The initial composition-induced wage response is $\mathcal{F}^{indirect}$. It induces a within-cohort worker reallocation of $\mathcal{E} v_{t+1}$, which feeds back into the relative wage through $\mathcal{H}$. In contrast, $\mathcal{F}^{direct}$ changes aggregate headcounts by changing cohort weights and therefore does not enter the $\mathcal{E}$ - $\mathcal{H}$ feedback loop. Solving the relative-wage equation gives

$$v_{t+1} = \frac{\mathcal{F}^{indirect}}{1 - \mathcal{E} \mathcal{H}}.$$

Substituting this response into the headcount equation gives the total dynamic response:

$$\boxed{\frac{d \phi_{A,t+1}}{d \phi_{A,t}} = \mathcal{F}^{direct} - \frac{\phi_{A,t+1} \phi_{N,t+1} \mathcal{E} \mathcal{F}^{indirect}}{1 - \mathcal{E} \mathcal{H}}}.$$

The denominator $1 - \mathcal{EH}$ therefore applies only to the worker reallocation initiated by the composition-induced wage response.

Proof of Lemma 3 Recall, Lemma 3 states:

Holding wages fixed, future generations respond to past reallocation according to

$$0 = \mathcal{F}^{direct, \text{bench}} < \mathcal{F}^{direct, \text{IGP-}\kappa} = \mathcal{F}^{direct, \text{IGP-}\tau}.$$

Proof. In the benchmark economy, equation (1) gives $\phi_{A,t+1}(A) = \phi_{A,t+1}(N)$ and therefore $\mathcal{F}^{direct, \text{bench}} = 0$. In each IGP economy, ag-born workers are more likely to choose agriculture than nonag-born workers, so $\mathcal{F}^{direct} > 0$. The restriction that the two IGP economies have the same allocation gap implies:

$$0 = \mathcal{F}^{direct, \text{bench}} < \mathcal{F}^{direct, \text{IGP-}\kappa} = \mathcal{F}^{direct, \text{IGP-}\tau}.$$

□

Proof of Lemma 4 Recall, Lemma 4 states:

The relative wage in period $t + 1$ responds to past worker reallocation according to

$$\mathcal{F}^{indirect, \text{IGP-}\kappa} > \mathcal{F}^{indirect, \text{IGP-}\tau} > \mathcal{F}^{indirect, \text{bench}} = 0.$$

Proof. In the benchmark economy, parental sector does not affect either sector choice or effective labor supply, so $\mathcal{F}^{H_A} = \mathcal{F}^{H_N} = 0$. Therefore, $\mathcal{F}^{indirect, \text{bench}} = 0$. Impose homothetic demand, $\vartheta_A = \vartheta_N = 1$, as maintained in the statement of the lemma. The aggregate-income term in the expression for $\mathcal{F}^{indirect}$ then vanishes. In either IGP economy, an increase in the fraction of ag-born parents raises effective labor supply in agriculture relative to nonagriculture. The remaining relative-price term therefore implies $\mathcal{F}^{indirect} > 0$. Substituting equation (2) into $\mathcal{F}^{H_A}$ and $\mathcal{F}^{H_N}$ shows that inherited productivity strictly increases the composition-induced effective-labor difference relative to inherited immobility. Since the two IGP economies have the same allocation gap, it follows that:

$$\mathcal{F}^{indirect, \text{IGP-}\kappa} > \mathcal{F}^{indirect, \text{IGP-}\tau} > \mathcal{F}^{indirect, \text{bench}} = 0.$$

□

Proof of Proposition 2 Recall, Proposition 2 states:

Past worker reallocation propagates positively in the IGP- τ economy and does not propagate in the benchmark economy:

$$\left(\frac{d\phi_{A,t+1}}{d\phi_{A,t}} \right)^{\text{IGP-}\tau} > 0, \quad \left(\frac{d\phi_{A,t+1}}{d\phi_{A,t}} \right)^{\text{bench}} = 0.$$

In the IGP- κ economy, the derivative can be positive or negative depending on whether the direct composition effect or the induced relative-wage effect dominates.

Proof. The denominator $1 - \mathcal{E}\mathcal{H}$ is strictly positive because $\mathcal{E} > 0$ and $\mathcal{H} < 0$. In the benchmark economy, both $\mathcal{F}^{\text{direct}}$ and $\mathcal{F}^{\text{indirect}}$ equal zero, so $d\phi_{A,t+1}/d\phi_{A,t} = 0$. In the IGP- τ economy, let $q = (\gamma-1)/\gamma$ and use the stationary allocations $a_t = m_t + (1 - m_t)\Delta_t$ and $b_t = m_t(1 - \Delta_t)$ introduced above. With inherited immobility, equation (2) gives $h_{j,t+1}(k) = \bar{\gamma}\phi_{j,t+1}(k)^{-1/\gamma}$ in every parent-sector and destination-sector cell. Substituting these effective-labor supplies into the expressions for $\mathcal{F}^{\text{indirect}}$ and $\mathcal{H}$ yields

$$\phi_{A,t+1}\phi_{N,t+1}\mathcal{E}^{\text{IGP-}\tau}\mathcal{F}^{\text{indirect,IGP-}\tau} < (1 - \mathcal{E}^{\text{IGP-}\tau}\mathcal{H}^{\text{IGP-}\tau})\Delta_t.$$

The inequality is strict because x^q is strictly concave for $q \in (0, 1)$ and $a_t > b_t$. Since $\mathcal{F}^{\text{direct,IGP-}\tau} = \Delta_t$, the total dynamic response is therefore strictly positive.

Finally, the total-response expression implies that the IGP- κ response is negative if and only if

$$\phi_{A,t+1}\phi_{N,t+1}\mathcal{E}^{\text{IGP-}\kappa}\mathcal{F}^{\text{indirect,IGP-}\kappa} > (1 - \mathcal{E}^{\text{IGP-}\kappa}\mathcal{H}^{\text{IGP-}\kappa})\mathcal{F}^{\text{direct,IGP-}\kappa},$$

In this case, the induced relative-wage effect dominates the direct composition effect. If the inequality is reversed, the direct composition effect dominates and the derivative is positive; equality gives a zero response. Hence the model does not impose an unconditional sign on the derivative in the IGP- κ economy. □

A.3. Quantitative Implementation and Additional Results

This appendix documents the data, production-side calibration, model inversion, counterfactual construction, and channel decomposition used in the quantitative analysis.

Data and timing. The quantitative exercise covers 1880, 1900, 1920, and 1940. The period- t cohort consists of ag-born and nonag-born workers with shares $\phi_{A,t-1}$ and $1 - \phi_{A,t-1}$, respectively. Thus the agricultural employment share in 1860, $\phi_{A,0} = 0.665393$, determines cohort composition in 1880; thereafter, the observed aggregate share in one census determines the composition used for the next census. Table 14 describes the inputs and their sources. Aggregate and subgroup employment shares are constructed from the linked censuses as described in Section 1. Agricultural output shares, relative prices, and real GDP per capita are from [Dennis and İşcan \(2009\)](#), [Alvarez-Cuadrado and Poschke \(2011\)](#), and [Maddison \(1995\)](#), respectively. I normalize the latter two series to one in 1880.

Table 14: Data Used in the Quantitative Analysis

Variable	Description and use	Source or construction
$\phi_{A,0}$	Initial share of ag-born workers entering in 1880; determines initial cohort composition.	1860 agricultural employment share in the linked census sample.
$\phi_{A,t}$	Aggregate agricultural employment share; targeted by the inversion and used as the next period's ag-born share.	Linked census sample, 1880–1940.
$\phi_{A,t}(A),$ $\phi_{A,t}(N)$	Agricultural employment shares of ag-born and nonag-born workers; untargted validation moments.	Linked census sample, 1880–1940.
$\omega_{A,t}$	Agricultural output share; calibrates and evaluates the production-side parameters.	Dennis and İşcan (2009) .
$p_t =$ $p_{N,t}/p_{A,t}$	Price of nonagricultural relative to agricultural goods; targeted by the inversion.	Alvarez-Cuadrado and Poschke (2011) ; indexed to one in 1880.
$Q_t = Y_t/P_t$	Real GDP per capita; targeted by the inversion.	Maddison (1995) ; indexed to one in 1880.

Note: Employment measures use the sample and sector definitions described in Section 1.

Production-side calibration. I set $\sigma = 0.3$, normalize $\Omega_N = 1$ and $\vartheta_A = 1$, and choose

$$\Omega_A = \frac{\omega_{A,1880}}{1 - \omega_{A,1880}} = 0.449275$$

so that the model exactly matches the 1880 agricultural output share. For any $\vartheta_N > 1$, the implied share is

$$\widehat{\omega}_{A,t}(\vartheta_N) = \frac{R_t(\vartheta_N)}{1 + R_t(\vartheta_N)}, \quad R_t(\vartheta_N) = \Omega_A p_t^{\sigma-1} Q_t^{(1-\vartheta_N)(1-\sigma)},$$

where $p_t = p_{N,t}/p_{A,t}$ and $Q_t = Y_t/P_t$. I estimate

$$\widehat{\vartheta}_N = \arg \min_{\vartheta_N > 1} \sum_{t \in \{1900, 1920, 1940\}} [\omega_{A,t} - \widehat{\omega}_{A,t}(\vartheta_N)]^2,$$

obtaining $\widehat{\vartheta}_N = 4.630$. The optimization uses the transformation $\vartheta_N = 1 + \exp(u)$ and multiple starting values; the reported fit is the solution with the smallest sum of squared residuals. The external CLM robustness calibration instead sets $(\sigma, \vartheta_A, \vartheta_N) = (0.33, 1, 1/0.33)$, while retaining the same Ω_A and Ω_N normalizations.

Period-by-period model inversion. I now provide the complete inversion underlying Proposition 3.

Step 1: Worker allocations and effective labor. Combining equations (1) and (3) implies that the product of the relative payoff shifter and relative wage must satisfy the following equation.

$$\phi_{A,t} = \frac{\phi_{A,t-1}}{1 + \left[\frac{B_{N,t}}{B_{A,t}} \frac{w_{N,t}}{w_{A,t}} \frac{1 - \tau}{\kappa_A} \right]^\gamma} + \frac{1 - \phi_{A,t-1}}{1 + \left[\frac{B_{N,t}}{B_{A,t}} \frac{w_{N,t}}{w_{A,t}} \kappa_N \right]^\gamma}. \quad (13)$$

The right-hand side of equation (13) is continuous and strictly decreasing in the positive product $(B_{N,t}/B_{A,t})(w_{N,t}/w_{A,t})$, with limits one as the product approaches zero and zero as the product approaches infinity. Therefore, every observed $\phi_{A,t} \in (0, 1)$ uniquely identifies this product and the two subgroup employment shares $\phi_{A,t}(A)$ and $\phi_{A,t}(N)$. Equations (2) and (4) then identify the following effective

labor supplies.

$$H_{A,t} = \bar{\gamma} \left[\phi_{A,t-1} \kappa_A \phi_{A,t}(A)^{(\gamma-1)/\gamma} + (1 - \phi_{A,t-1}) \phi_{A,t}(N)^{(\gamma-1)/\gamma} \right], \quad (14)$$

$$H_{N,t} = \bar{\gamma} \left[\phi_{A,t-1} (1 - \phi_{A,t}(A))^{(\gamma-1)/\gamma} + (1 - \phi_{A,t-1}) \kappa_N (1 - \phi_{A,t}(N))^{(\gamma-1)/\gamma} \right]. \quad (15)$$

Both effective labor supplies are finite and strictly positive because $\gamma > 1$ and the subgroup employment shares are interior.

Step 2: Relative technology. Market clearing and sectoral production imply $X_{j,t} = Z_{j,t} H_{j,t}$. Substituting sectoral production into equation (10) gives the following condition.

$$\frac{Z_{A,t} H_{A,t}}{Z_{N,t} H_{N,t}} = \frac{\Omega_A}{\Omega_N} p_t^\sigma Q_t^{(\vartheta_A - \vartheta_N)(1-\sigma)}. \quad (16)$$

Rearranging this condition uniquely identifies the relative technology.

$$\frac{Z_{N,t}}{Z_{A,t}} = \frac{H_{A,t}}{H_{N,t}} \frac{\Omega_N}{\Omega_A} p_t^{-\sigma} Q_t^{(\vartheta_N - \vartheta_A)(1-\sigma)}. \quad (17)$$

Step 3: Technology levels. Using market clearing, the identified relative technology, and the observed Q_t in equation (8), I recover the sectoral technologies as follows.

$$Z_{A,t} = \left[\Omega_A^{1/\sigma} \left(\frac{H_{A,t}}{Q_t^{\vartheta_A}} \right)^{(\sigma-1)/\sigma} + \Omega_N^{1/\sigma} \left(\frac{(Z_{N,t}/Z_{A,t}) H_{N,t}}{Q_t^{\vartheta_N}} \right)^{(\sigma-1)/\sigma} \right]^{-\sigma/(\sigma-1)}, \quad (18)$$

$$Z_{N,t} = \frac{Z_{N,t}}{Z_{A,t}} Z_{A,t}. \quad (19)$$

The expression in brackets is finite and strictly positive, so both recovered technologies are finite, strictly positive, and unique.

Step 4: Relative payoff shifter. Under the agricultural-good numeraire, equation (7) determines the relative wage.

$$\frac{w_{N,t}}{w_{A,t}} = p_t \frac{Z_{N,t}}{Z_{A,t}}.$$

Combining the identified relative wage with the nonag-born employment share in equation (1) identifies the relative payoff shifter.

$$\frac{B_{N,t}}{B_{A,t}} = \left(\frac{1 - \phi_{A,t}(N)}{\phi_{A,t}(N)} \right)^{1/\gamma} \frac{1}{\kappa_N p_t(Z_{N,t}/Z_{A,t})}. \quad (20)$$

Only the relative payoff shifter is identified because multiplying $B_{A,t}$ and $B_{N,t}$ by a common positive constant does not change any worker's sector choice.

Proof of Proposition 3. Equation (13) uniquely identifies the product of the relative payoff shifter and relative wage, which identifies the subgroup employment shares and the positive effective labor supplies in equations (14) and (15). Equation (17) then uniquely identifies the relative technology. Given the relative technology, equations (18) and (19) uniquely identify both technology levels. Finally, equation (20) uniquely identifies the relative payoff shifter. Each mapping is single-valued and maps the stated interior observables into strictly positive fundamentals, establishing the result. $\square$

Recursive implementation and model fit. Starting from the observed or imposed initial condition $\phi_{A,0}$, I apply the inversion sequentially because the recovered allocation in period t supplies the inherited composition $\phi_{A,t}$ used in period $t + 1$. In the historical inversion, I use the observed $\phi_{A,t}$, rather than its fitted value, as the next period's inherited composition. At every date, the numerical implementation reconstructs the three targeted outcomes and the CES aggregator with absolute errors below 10^{-8} . The agricultural output share is not an additional identifying moment in this inversion. Once p_t and Q_t are imposed, equation (11) determines the agricultural output share from the calibrated production-side parameters. The discrepancy from the observed output share therefore remains the production-side calibration residual rather than a fourth equation in the period-by-period inversion.

Untargeted subgroup employment shares. The inversion also implies $\phi_{A,t}(A)$ and $\phi_{A,t}(N)$, but does not target their observed counterparts. Conditional on $\phi_{A,t-1}$ and the time-invariant labor-supply parameters, the single aggregate target $\phi_{A,t}$ identifies the common payoff-wage term in equation (13), which in turn fixes both subgroup shares. Their comparison with the data in Figure 9 is therefore an over-identifying check. The labor-supply wedges are estimated from 1940 income and

sector-choice data, so the model does not reproduce the substantially greater subgroup persistence observed in 1880. Matching these subgroup paths would require time-varying wedges or an additional model mechanism.

Counterfactual IGP regimes. Let superscript b denote the benchmark inversion and define the benchmark growth factors

$$g_t^A = \frac{Z_{A,t}^b}{Z_{A,1880}^b}, \quad g_t^N = \frac{Z_{N,t}^b}{Z_{N,1880}^b}, \quad g_t^B = \frac{(B_N/B_A)_t^b}{(B_N/B_A)_{1880}^b}.$$

I consider the benchmark regime, no inherited productivity ($\kappa_A = \kappa_N = 1$), no inherited immobility ($\tau = 0$), and no IGP ($\kappa_A = \kappa_N = 1, \tau = 0$). For each regime r , I first apply the inversion under that regime's parameters to the common 1880 targets. This produces regime-specific anchors $Z_{A,1880}^r$, $Z_{N,1880}^r$, and $(B_N/B_A)_{1880}^r$. The subsequent fundamentals are

$$Z_{A,t}^r = Z_{A,1880}^r g_t^A, \quad Z_{N,t}^r = Z_{N,1880}^r g_t^N, \quad (B_N/B_A)_t^r = (B_N/B_A)_{1880}^r g_t^B.$$

I then solve the equilibrium forward, using the simulated $\phi_{A,t}^r$ as the parental composition of the next cohort. This construction makes all regimes match the 1880 agricultural employment share, relative price, and real GDP per capita, while exposing them to the same proportional growth in fundamentals thereafter. Worker reallocation is $100(\phi_{A,1880}^r - \phi_{A,1940}^r)$ percentage points.

Interval-level Shapley decomposition. For regime r and an interval ending in t , let F_0^r and F_1^r denote the beginning- and end-of-interval fundamentals, and let M_0^r and M_1^r denote the corresponding inherited parental compositions. Write

$$y_{ab}^r = \phi_A(F_a^r, M_b^r), \quad a, b \in \{0, 1\}.$$

The two-player Shapley contributions to the decline in the agricultural employment share are

$$C_t^r = \frac{1}{2} [(y_{00}^r - y_{10}^r) + (y_{01}^r - y_{11}^r)],$$

$$P_t^r = \frac{1}{2} [(y_{00}^r - y_{01}^r) + (y_{10}^r - y_{11}^r)],$$

where C_t^r is the contemporaneous response to technologies and the relative payoff shifter and P_t^r is intergenerational propagation through parental composition. The averages span the two possible orders in which the inputs change. They satisfy

$$C_t^r + P_t^r = y_{00}^r - y_{11}^r$$

in every interval. I sum each channel over 1880–1900, 1900–1920, and 1920–1940, then report its counterfactual-minus-benchmark difference in percentage points:

$$100 \sum_t (C_t^r - C_t^b), \quad 100 \sum_t (P_t^r - P_t^b).$$

Their sum equals the difference between counterfactual and benchmark worker reallocation reported in Table 6.

Robustness implementation. Each robustness exercise repeats the entire procedure: re-invert the historical fundamentals under the alternative parameters, re-anchor every IGP regime to the common 1880 targets, simulate forward using that specification’s inferred growth factors, and recompute the interval-level Shapley decomposition. The CLM exercise changes only the production-side parameters to $(\sigma, \vartheta_A, \vartheta_N) = (0.33, 1, 1/0.33)$. The added-fixed-effect labor-supply exercise retains the baseline production parameters and uses

$$(\gamma, \kappa_A, \kappa_N, \tau) = (2.293, 1.205, 1.431, 0.341).$$

The topcode-adjusted exercise likewise retains the baseline production parameters and uses

$$(\gamma, \kappa_A, \kappa_N, \tau) = (2.063, 1.434, 1.809, 0.401).$$

Thus the robustness results incorporate both the direct effect of alternative parameters and the associated changes in the inferred historical fundamentals; they do not hold the benchmark fundamental paths fixed in levels or growth rates.